An Informational Theory of Privacy*

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Abstract

We develop a theory that explains how and when privacy can increase welfare. Without privacy, some individuals misrepresent their preferences, because they will otherwise be statistically discriminated against. This "chilling effect" hurts them individually, and impairs information aggregation. The information gain from infringing privacy (e.g. by electronic surveillance) can be much smaller than expected ceteris paribus. Overall, privacy is essential for any mechanism of information aggregation, such as markets or a democratic society. It is also redistributive: Like free speech, privacy benefits some and hurts others.

JEL: D72, D82, J71, K40

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Privacy is one of the most pressing issues of the information age. It is at the center of debates about government response to extremism and terrorism, especially after revelations that many western governments systematically infringe the privacy of their own citizens by engaging in indiscriminate electronic surveillance (cf. Schneier (2015), Greenwald (2014) or Economist (2013)). The fact that people are willing to give up some privacy in exchange for lower prices or better services is part of the business model of many companies. Some of the most successful and fastest-growing businesses are even built on the fact that people are willing to trade personal data in exchange for a free service, thus turning their data into a product which can then be profitably sold.

As these examples show, issues of privacy often involve trade-offs: Between privacy and security, privacy and thriftiness, or even privacy and participation in public life. Understanding the value and the effects of privacy is crucial for how voters, consumers

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and regulators approach these trade-offs. Classical economic theory suggests that privacy is usually welfare-reducing because it creates asymmetric information – an idea that is echoed by probably the best-known economic treatise on privacy, Posner (1981).

In this paper, we develop an informational theory of privacy.[1] By considering the role of privacy in allowing individuals to express their preferences and in information aggregation, we show that privacy can enhance, not reduce, welfare. One part of this insight is that as privacy allows people to express their preferences and opinions more freely, it can actually improve overall information aggregation of a society. Moreover, the welfare gains from infringing privacy, such as better information about individuals, are often not as large as the losses of the individuals, because individuals will react to a loss of privacy by changing their behavior and thus providing less information. Privacy can also protect minorities: Those with opinions or preferences which are different from the median of the population.

To illustrate our results, consider the following example. Alice would prefer if marijuana was legalized. She considers publishing an overview of her arguments on a social network to try to convince her friends. However, we assume that in Alice's world there is very little privacy: If she does something online, everyone can see it – not just her friends, but also future employers, her parents, the police, and so on.

There is some correlation between preferences on legalization and actual drug use: people who actually use drugs are more likely to support legalization. The correlation is of course far from perfect – many people might support legalization for philosophical or practical reasons without being users, and some drug users might even oppose it.

Employers do not want to hire drug users, but drug use is not observable. An employer will therefore use the observable characteristic (whether Alice did or did not publicly support legalization) to make a hiring decision: People who have supported legalization will not be hired. We can show that this happens in equilibrium if the correlation between types (i.e. drug use and preference for legalization) is sufficiently high. Being unable to observe the attribute that he is really interested in, the employer will statistically discriminate (as in Arrow (1973) and Phelps (1972)) based on observed preference.

Then, however, Alice has to make a choice: Voice her preference, and risk not being hired for a job – or stay quiet, and face no such consequence. If she doesn't feel strongly about the subject (i.e. if she only has a weak preference for legalization), she will choose not to express her opinion. That is our first result: Lack of privacy causes a "chilling effect". Despite Alice's preference being not only legal and legitimate, but also insubstantial for the job (recall that even the employer does not take issue with her preference itself), she decides not to express it for fear of the consequence.[2]

[1] By "privacy", we mean the ability to take actions without being observed, and having interactions with others confined to the intended recipients. This is only one of many possible definitions and understandings of the term "privacy"; see for example Solove (2010) for an overview.

[2] The term "chilling effect" has been used by legal scholars since at least 1952, when U.S. Supreme

If Alice had been an opponent of legalization, there would have been no chilling effect. The spectrum of expressed preferences that are present in the public debate will therefore be skewed: Those who oppose legislation speak out freely, while those who support it tend to stay quiet. Since the optimal policy should be an unbiased aggregation of individual preferences, the policy that is implemented will systematically deviate from this optimum.

If Alice's views on the matter are strong, however, she decides to post her arguments anyway since the expected gain from doing so outweighs the disadvantage of not being hired. In this case the lack of privacy hurts Alice in two ways. She suffers from not being hired and – due to the chilling of others with more moderate opinions – her preferred policy of legalization is less likely to be implemented. This is our second result: Lack of privacy hurts those with non-mainstream preferences the most – those that care too strongly to protect themselves by adapting their behavior (i.e. giving in to the chilling effect). Again, note that this effect is asymmetric, too, and affects only those who strongly support legalization.

The change in behavior that results from the chilling effect also has another consequence: It makes the employer's statistical discrimination less effective. Since many people (both drug users and non-users) now misrepresent their preferences, observing what someone posts about drug legalization can become less informative. This is true in particular if actual drug users are more afraid of not being hired, say because of worse outside options, than non-users. In this case, actual drug users are chilled more than non-users, and drug use becomes relatively less likely among those who still decide to support legalization. This is our third result, which is especially important in debates about government surveillance. Instead of an "employer", think for a moment that it was the police who decides to watch those closer who voice a certain opinion, or use certain means of communication, or visit certain places. Surveillance programs and other privacy intrusions should therefore never be naively evaluated *ex ante* – they will always be made less effective by the resulting shift in behavior. Herein lies the crux of our analysis of the trade-offs involved in privacy – that we don't just need to weigh the loss of those who lose privacy with the gain of those who gain information, but also consider how large that gain actually is, given that people rationally change their behavior if they know they are being observed. Such changes in behavior happen: Martews and Tucker (2015) show a significant shift in search engine search terms after the Snowden revelations in 2013; a survey of American writers has found that 1 in 6 has avoided writing or speaking on a particular topic for fear of surveillance (PEN America, 2013).

Still: Even if society in general and some applicants in particular are worse off without privacy, surely we would at least expect the employer to gain from being able to make a distinction between applicants according to how likely they are to be drug users? We show that even this is far from guaranteed. Under some conditions, the chilling effect can

Court Justice Felix Frankfurter used it in a concurring opinion in Wieman v. Updegraff, 344 U.S. 183.

be so strong that in equilibrium, the employer is indifferent between being able or not being able to observe whether people support legalization. This happens, for example, in large populations in which individuals care more about the potential losses of being discriminated against than the unlikely case of being able to make a difference by expressing their opinion. While it is still rational for the employer to use the information if he has it, he does not gain from doing so in equilibrium and society as a whole loses. This is our fourth result.

We also consider the case in which Alice derives intrinsic utility from speaking her mind (instead of trying to influence a public policy decision). Since this intrinsic utility is independent of other people's actions, it puts an upper limit on the magnitude of the chilling effect. While this guarantees that the employer can derive modest gains from being able to observe applicants' preferences, we show that privacy is still welfare-optimal unless both the correlation between opinion and drug use and the per-person gain from information for the employer are sufficiently high.

In an extension, we ask: What if Alice herself can choose to keep her message private? Then the employer has to treat all applicants about whom he can find no information equally. Would he hire someone about whom he can find no information? That depends on how people who oppose legalization behave: Would they also choose to keep their messages private? Intuitively, they have nothing to fear from not doing so. We argue that a possible equilibrium where everybody, supporters and opponents of legalization, individually choose privacy for themselves is not stable. Another equilibrium, in which only supporters of legalization choose privacy and in which privacy therefore is meaningless, is more stable. To work well, privacy can therefore not always be left to the individual – sometimes it needs to be mandated.[3]

Finally, in our second extension, we look more closely at Alice's rational response to having no privacy. We have established above that Alice might misrepresent her opinion if she prefers legalization (the "chilling effect"), but that she will otherwise express herself truthfully if she strongly supports legalization. If she does so, however, she is aware that this will come with negative consequences, such as worse employment prospects. Now, assume that Alice can take a costly action to mitigate the damage to herself – however, this is costly to the employer. In our example, imagine that Alice could bring a lawyer to every job interview – which slightly decreases, ceteris paribus, the probability of not being hired because of her expressed opinions (thus making it her rational response to the employer's statistical discrimination), while making the interaction much more cumbersome for the employer. In this way, the statistical discrimination that results from lack of privacy can erode trust among the players, and can mean that by a chain of rational responses to each

[3]There are parallels to the obligatory secret ballot, consider for example the point made by Schelling (1960): If ballot secrecy was optional, voters could be intimidated into making their ballot public. Forbidding them from doing so protects them from any such intimidation.

other's behavior, they end up in a Pareto-inferior equilibrium.

Our general model, which we introduce in section 1, considers a problem of information aggregation, in which a group of individuals have cardinal preferences over two options and express their preference by supporting one of the two options. We do not restrict our arguments to any specific information aggregation problem – in fact, our only assumptions about the information aggregation mechanism are that the probability of an option being implemented increases in the number of supporters that it has, and that the process is not systematically biased towards one option. Our model is therefore applicable to a large variety of situations. The example above already points to political information aggregation through public debate or voting. However, the mechanism might just as well be a market in which two providers of goods or services compete for customers. Efficiency demands that the provider who is preferred by most customers also does more business. But if using one of the providers is in some way disreputable, or can bring adverse consequences, lack of privacy and the chilling effect will systematically bias the result. We discuss examples of the mechanism in section 6.

What kind of privacy problem do we have in mind when we assume, as in our example, that some observable behavior is predictive of an unobservable type? Here, too, we keep our assumptions quite general, as we only assume that the two unobservable types (in our example: policy preference and drug use) are positively correlated. It is crucial to note that this does not require any sort of causal relationship – only correlation. We think that in the real world, almost any variable can be "predictive", in the sense of our model, of almost any other variable. Meehl (1990) calls this the "crud factor" and notes that "in social science, everything is somewhat correlated with everything." Even minor choices are correlated with political preferences – a fact which is being used by many political parties and candidates to identify their potential voters. Hamburger and Wallsten (2005), for example, report about microtargeting efforts by the Republican party in the United States that "... bourbon drinkers were more likely to be Republican, while gin was a Democrat's drink. ... Democrats preferred Volvos; Ford and Chevy owners were most likely Republicans. People with call-waiting service on their telephones were predominantly Republican." As the availability of data and of cheap processing power has grown substantially in recent years, this has become much more acute. The consulting firm Deloitte advertises that it can reliably predict people's life expectancy from observing their buying decisions (Robinson et al., 2014, p. 6), and the "big data underwriting firm" Zest Finance simply claims: "All Data is Credit Data."

We would also like to point out that the assumption that we make about correlated observables and unobservables applies in almost all contexts, economic or otherwise. It is quite rare that banks, employers, law enforcement agencies or indeed anyone else can directly and unambiguously observe the variable that they are really interested in. One's future financial situation or intellectual ability, whether one is a terrorist, a criminal or

a reliable friend are all in essence unobservable. Through years of everyday experience, we have gotten used to forming estimates through statistical discrimination by using (multiple) observables. But all observation is ultimately incomplete, and the correlation between what we conclude based on our observations and the truth is never 100 percent. A police officer could be trying to judge whether someone carries a gun from a short distance based on what he sees in the suspect's hand, or whether someone is planning a terrorist attack based on their internet search history: the difference in correlation between the two situations is quantitative, not qualitative.

In understanding privacy as the creation and maintenance of asymmetric information, our study takes a similar point of departure as the "Chicago school", exemplified by Stigler (1980) and Posner (1981). However, they go on to argue that since asymmetric information creates economic inefficiencies and reduces welfare, privacy must be welfare-reducing. This line of thought echoes the ubiquitous "nothing to hide"-argument, which Schneier (2006) has called "the most common retort against privacy advocates." As Solove (2010) points out, this argument usually takes the form: "If you aren't doing anything wrong, what do you have to hide? ... If you have nothing to hide, what do you have to fear?"

Our model allows us to argue that this argument, and hence the claim that privacy necessarily reduces welfare, is based on two faulty assumptions. Firstly, it assumes that all information is precise and unambiguous. But decisions that are made under uncertainty are routinely based on statistical discrimination. Not everybody who travels to Yemen is doing so to attend a terrorist training camp; yet it might be rational for Western governments to watch people who undertake such travels more closely – to the detriment of someone who is planning to visit his family in Yemen.

The Chicago argument therefore offers only limited guidance when it comes to actual problems of privacy. It is plausible that the first-best could be achieved in the total absence of asymmetric information. But in the real world, asymmetric information is a fact of life, and questions of privacy are therefore about *how much* asymmetric information there should be, and how it should be structured. The Chicago argument addresses an imaginary ideal case and has little to say about intermediate cases (and whether, for example, welfare is monotone in the amount of asymmetric information), which limits its use substantially.[4]

Secondly, it takes a naive ex-ante view of rational behavior. People who know that their actions are being observed often optimally change their behavior. Since traveling to Yemen can make one the target of extensive surveillance, both terrorists and non-terrorists might choose to abstain from such travels. This is clearly welfare-reducing for the non-terrorist whose plans are disrupted, and not necessarily counterbalanced by a welfare gain from deterring the terrorists' travels (especially if he still gets his training but simply at

[4] A similar argument against the Chicago school is made by Hermalin and Katz (2006).

another place).[5]

Accepting our argument that privacy can be welfare-enhancing, and that sometimes privacy even needs to be mandated to work, also means refuting the Chicago argument that any regulation of privacy can at best be ineffective and at worst damaging.

Two recent papers have proposed rationales for privacy in public good settings where agents have an intrinsic motivation to contribute and also care about their image. That is, each agent would like others to believe that he has a high intrinsic motivation. Daughety and Reinganum (2010) show that privacy can be optimal in this setting if a lack of privacy would lead to excessive contributions due to image concerns. Ali and Bénabou (2016) add a principal who has to decide on his own contribution in a setting where agents and principal have only noisy information about the usefulness of the public good. More privacy implies that the aggregate contribution by the agents is – as a signal of the usefulness of the public good – more informative and therefore allows the principal to better choose his contribution. The mechanism in our model differs in two important ways: First, we do not rely on image concerns but microfound the downside of taking a certain action (e.g. supporting drug legalization) through an interaction with another player (e.g. a future employer). Note that image concerns are not a reduced form for this because the (changing) utility of the interacting player is an integral and indispensable part of our welfare analysis.[6] Second, the inference is somewhat more subtle in our model as the interacting player is not interested in the preference for action (e.g. the preference for drug legalization) but only in unobservables that are correlated with this preference (e.g. drug use). In this sense, we link the literature on statistical discrimination (Arrow, 1973; Phelps, 1972) and the literature on privacy.

Apart from such general economic studies of privacy, there is a large literature in industrial organization and related fields that deals with demand for privacy and the meaning of privacy for issues like pricing. Acquisti (2010) and Acquisti et al. (2015) provide excellent overviews; here we want to point to some studies that are loosely related to ours.

Hirshleifer (1971) argues that information revelation before trading can impair risk-sharing and therefore reduce welfare. This "Hirshleifer effect" means, for example, that providing health data about buyers of life insurance transfers risk from the seller to the more risk-averse buyers. This can be understood as an argument for privacy. Hermalin and Katz (2006) follow in a similar vein and show that privacy can be efficient in a model of price discrimination by a monopolist and a model of a competitive labor market. They

[5] Cummings et al. (2015), who analyze a model in which a consumer reveals information to an advertiser by his buying decision, make a similar argument that the *equilibrium effects* of privacy are different from what we would expect in a static setup.

[6] Morris (2001), in a model of "political correctness," derives image concerns from model primitives in a way that has similarities to our model, but does not carry out a welfare analysis beyond identifying several countervailing effects.

also show that allocating property rights to control information does not affect equilibrium outcomes (and therefore the results) in their setup. Calzolari and Pavan (2006) consider information exchange between principals who contract with the same agent, and find that privacy can increase welfare.

Similar to the second extension of our model, Acquisti and Varian (2005) consider rational reactions by people who lack privacy – for example, that internet users employ anonymization tools. They argue that this can make it unprofitable (and hence inefficient) for the seller of goods to collect information.

1. Model

The model has two stages. First, an information aggregation stage in which each of n individuals has to decide between two options. Second an interaction stage in which each individual interacts with an opposing player (OP).

In the information aggregation stage, each of n individuals has to choose to support one of two options, which are either $p = 1$ or $p = 0$.[7] Individual i's choice is denoted by $p_i \in \{0, 1\}$. If m individuals choose $p_i = 1$, the probability that $p = 1$ is $q(m/n)$. We assume that q is continuously differentiable and strictly increasing in m/n, that is, each option is more likely to be implemented the higher the proportion of individuals who chose it.[8] We also assume that q is not systematically biased towards one of the two options, i.e. we assume that it is point-symmetric in its argument around 0.5. The payoff of $p \in \{0, 1\}$ for individual i is then $\theta_i p$. Individual i's valuation θ_i is privately known by i. We assume that the θ_is are iid draws from a standard normal distribution Φ.

Before describing the interaction stage let us give an example for the information aggregation stage.

Example 1. *There is a petition to liberalize drug laws to a certain degree. The more citizens sign the petition, the more likely it is that its demands will be implemented. Every citizen has to decide whether to sign the petition ($p_i = 1$) or not ($p_i = 0$). Every citizen has an expected payoff consequence of liberalization of θ_i.*

We now turn to the interaction stage. In this stage, each individual interacts with one opposing player (OP). We will describe this player as one central outside player, although nothing in the model rules out the alternative case where each individual interacts with a different player (possibly even one of the other individuals). OP has to choose how he interacts with individual i and he can choose from the actions A (aggressive) or M (mild). We normalize OP's payoff from playing M to 0 and assume that the payoff of playing

[7] In the supplementary material to the paper, we show that our results are robust to giving individuals the possibility to "abstain".

[8] Differentiability will allow us to later analyze the effect of large n in proposition 3 – it has no effect on our other results.

A against a type τ_i is simply τ_i which is a private characteristic of individual i. The characteristics τ_i are drawn independently from a distribution Γ_{θ_i} with support in $[\underline{\tau}, \bar{\tau}]$. We assume that $\Gamma_{\theta_i'}$ first order stochastically dominates $\Gamma_{\theta_i''}$ if and only if $\theta_i' \geq \theta_i''$. This implies that θ_i and τ_i are positively correlated as higher θ_i make higher τ_i more likely. We also assume that $\Gamma_\infty = \lim_{\theta_i \to \infty} \Gamma_{\theta_i}$ is a non-degenerate distribution in the sense that $\Gamma_\infty(\tau_i) > 0$ for all $\tau_i > \underline{\tau}$ – a technical property that will be useful for some of our welfare results.

To make the problem interesting, we assume that A is OP's best response if $\tau_i = \bar{\tau}$ and M is the best response if $\tau_i = \underline{\tau}$. That is, $\bar{\tau} > 0$ and $\underline{\tau} < 0$. Furthermore, we assume that $\mathbb{E}[\tau_i] \leq 0$, so that M is an optimal response to any individual about whom nothing is known. However, OP does not observe τ_i when choosing his action and will only be able to form expectations about the individual's τ_i. We will distinguish two cases: In the *privacy* case, we consider OP's problem when he has no information on τ_i apart from the priors Γ_{θ_i} and Φ; in particular OP does not know p_i in this case. Our above assumption that $\mathbb{E}[\tau_i] \leq 0$ means that in this case, the OP's best response is to play M against all individuals since the expected payoff of playing A against any individual is simply $\mathbb{E}[\tau_i]$.

The more complicated part of the analysis, however, will deal with the case *without privacy* in which OP observes which opinion i voiced in the information aggregation stage, i.e. OP can observe p_i and can condition his expectation of τ_i on this information. The individual's payoff is normalized to 0 when OP plays M. If OP plays A against i, then i will have a payoff of $-\delta(\tau_i)$ where $\delta > 0$ is a differentiable function that is weakly increasing in τ_i. We assume that individual payoffs from the two stages are additive. All players are assumed to maximize their expected payoff.[9]

Figure 1 shows a graph of the model which illustrates the two types that each individual has, and how they are correlated. We will use and modify this figure in the following sections to illustrate our main points. Table 1 shows the payoffs to the OP and an individual from a single interaction to illustrate the trade-off that OP faces between treating players who chose $p_i = 1$ aggressively or mildly.

		Individual chooses	
		$p_i = 1$	$p_i = 0$
OP treats $p_i = 1$ ...	... aggressively (A)	$\tau_i, p\theta_i - \delta(\tau_i)$	$0, p\theta_i$
	... mildly (M)	$0, p\theta_i$	$0, p\theta_i$

Table 1: The payoffs of OP and a single individual under the assumption that OP treats all individuals who choose $p_i = 0$ mildly. The OP interacts with all individuals. His payoff is the sum of the payoffs from all these interactions, but his choice of how to treat individuals applies to all interactions. Note that the payoff of the individual depends on p, which he in turn influences with his choice of p_i.

[9]For the main part of the paper, we assume that q is the same across the two cases– privacy and no privacy. Section 5.1 endogenizes q and shows that our results are robust to relaxing this assumption.

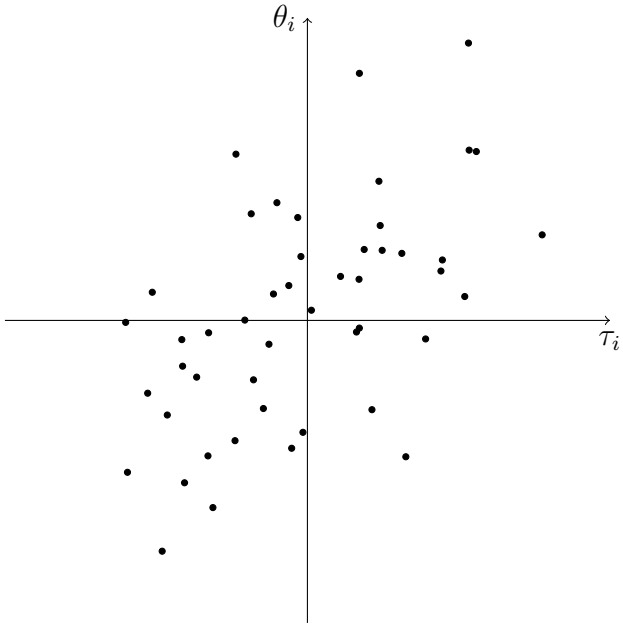

Figure 1: An illustration of our model. Each dot represents an individual. Individual i's type τ_i and θ_i are positively correlated. The OP wants to treat those with $\tau_i > 0$ aggressively and all others mildly, but he cannot observe τ_i. Individuals' preferences are given by θ_i; observing the choices of individuals will therefore provide the OP with information about τ_i. "Privacy" is the question whether the OP can or cannot observe an individual's choice before deciding how to treat her.

Example 1 (Continued)**.** *Continuing our example, OP might be a potential employer who has to decide whether to hire citizen i (action M) or not to hire i (action A). The employer would prefer to hire i if i is not a drug user and would prefer not to hire i if i is a drug user. The type τ_i would then be binary, i.e. $\tau_i \in \{\underline{\tau}, \bar{\tau}\}$, and would indicate whether i is a drug user or not. The first order stochastic dominance assumption on Γ_{θ_i} then simply means that the probability of being a drug user is increasing in θ_i. Hence, τ_i and θ_i are positively correlated which also means that citizens who support drug legalization are relatively more likely to be drug users than citizens opposing legalization. Citizen i prefers to be hired and the disutility of not being hired – denoted by δ – might be bigger for drug users because their outside options are generally worse.*

2. Preliminary Analysis – The Chilling Effect

2.1. OP's Beliefs

We start the analysis with some preliminary results on the individuals' and OP's beliefs and strategies. This will then allow us to establish the chilling effect and analyze its welfare implications.

The payoff of individual i from the information aggregation stage is $p * \theta_i$. The higher θ_i, the higher is i's benefit from $p = 1$. Given this structure, it is not surprising that i will use a cutoff strategy: If θ_i is higher than some cutoff/threshold $t(\tau_i)$, i chooses $p_i = 1$ and

otherwise he chooses $p_i = 0$. In the privacy case, payoffs of the interaction stage do not depend on actions chosen in the information aggregation stage and therefore i will choose $p_i = 1$ if and only if θ_i is positive. This pins down the equilibrium of the privacy case as we already established that OP plays M there by $\mathbb{E}[\tau_i] \leq 0$.

Lemma 1. *Only cutoff strategies are rationalizable for individuals, i.e. each individual will choose a cutoff $t(\tau_i)$ and play $p_i = 0$ if $\theta_i < t(\tau_i)$ and $p_i = 1$ if $\theta_i > t(\tau_i)$. In the privacy case, the optimal cutoff $t^p(\tau_i) = 0$.*

Given a cutoff strategy $t(\tau_i)$, we can determine the beliefs of OP in the case without privacy using Bayes' rule as

$$\beta_1(\tau) \equiv prob(\tau_i \leq \tau | p_i = 1) = \frac{\int_{\mathbb{R}} \int_{\underline{\tau}}^{\tau} \mathbb{1}_{t(\tau_i) \leq \theta_i} \, d\Gamma_{\theta_i}(\tau_i) \, d\Phi(\theta_i)}{\int_{\mathbb{R}} \int_{\underline{\tau}}^{\bar{\tau}} \mathbb{1}_{t(\tau_i) \leq \theta_i} \, d\Gamma_{\theta_i}(\tau_i) \, d\Phi(\theta_i)} \tag{1}$$

$$\beta_0(\tau) \equiv prob(\tau_i \leq \tau | p_i = 0) = \frac{\int_{\mathbb{R}} \int_{\underline{\tau}}^{\tau} \mathbb{1}_{t(\tau_i) \geq \theta_i} \, d\Gamma_{\theta_i}(\tau_i) \, d\Phi(\theta_i)}{\int_{\mathbb{R}} \int_{\underline{\tau}}^{\bar{\tau}} \mathbb{1}_{t(\tau_i) \geq \theta_i} \, d\Gamma_{\theta_i}(\tau_i) \, d\Phi(\theta_i)}. \tag{2}$$

That is, $\beta_1(\tau)$ is the probability that τ_i is below τ given that i chose $p_i = 1$. These beliefs allow us to define OP's expected utility of playing A conditional on observing decision p_i and given cutoff strategy $t(\tau_i)$:

$$v_1 = \int_{\mathbb{R}} \tau \, d\beta_1(\tau) \tag{3}$$

$$v_0 = \int_{\mathbb{R}} \tau \, d\beta_0(\tau). \tag{4}$$

The best response of OP to a given cutoff strategy is to choose A against an individual who chose $p_i = j$ if $v_j > 0$ for $j \in \{0, 1\}$. Otherwise, it is a best response to choose M.[10]

2.2. The Chilling Effect

For the case without privacy, the following lemma states that OP is more likely to play A against individuals who have chosen $p_i = 1$ in the information aggregation stage than against those who have chosen $p_i = 0$. Intuitively, individuals with a high θ_i have more to gain from choosing $p_i = 1$ in the information aggregation stage. As θ_i and τ_i are positively correlated, OP is relatively more likely to play A against them.

Lemma 2. *In every perfect Bayesian equilibrium, $v_1 \geq v_0$.*

The previous lemma is the basis of the chilling effect. In equilibrium, OP is more likely to play A against individual i if i chose $p_i = 1$ in the information aggregation stage.

[10] Note that OP's best response does not depend on the number of individuals choosing $p_i = 1$ in the first stage. Intuitively, this information does not contain any information about τ_i (given that p_i is known) because all θ_i and τ_i are independently drawn by assumption.

For this reason, i is to some degree afraid of choosing $p_i = 1$. More technically, there are types (θ_i, τ_i) for which an individual would choose $p_i = 1$ in the privacy case but would choose $p_i = 0$ if OP learns p_i before taking his action. The decision in the information aggregation stage is therefore biased against $p = 1$ in the case without privacy.

There is one minor caveat to this result: If OP's preferences are so strong that he always uses the same action, e.g. OP prefers to play M against individuals who have played $p_i = 0$ and individuals who have played $p_i = 1$, then no chilling occurs because information on p_i is not relevant for OP's decision and the equilibria with and without privacy are identical. Put differently, chilling occurs whenever information about p_i matters for OP's behavior but cannot occur if this information does not affect OP's behavior.

Proposition 1 (Chilling effect)**.** *The equilibrium cutoff for every type τ_i is weakly higher without privacy than in the privacy case. The inequality is strict whenever the absence of privacy changes the equilibrium behavior of OP. The difference of equilibrium cutoff without and with privacy is increasing in τ_i.*

Figure 2 illustrates the chilling effect. The effect changes the behavior of individuals with moderate preferences – that is, those who are almost indifferent between the two options – as it simply shifts the cutoff upwards by a bit. Individuals with a very high preference for $p = 1$ will choose $p_i = 1$ with and without privacy and individuals with a very low (that is, negative) preference will choose $p_i = 0$ in both cases. Those that are almost indifferent but choose $p_i = 1$ in the privacy case are the ones who change their behavior when OP uses information about p_i. In this sense, the individuals who change their behavior do not lose a lot by their behavior change. However, individuals with strong preferences for $p = 1$ should be most worried about chilling: They do not change their own behavior but – because chilling changes the behavior of those with more moderate preferences – $p = 1$ will be less likely without privacy than it would have been with privacy. In short, privacy changes the behavior of moderate people and protects people with extreme preferences.

The result that the cutoff is shifting more for individuals with a high type τ_i implies that the cutoffs of higher τ_i are higher. As a consequence, abolishing privacy becomes somewhat less profitable for OP compared to the case where individuals use the same cutoff: The fact that higher τ_i have higher cutoffs reduces the correlation between θ_i and τ_i. This correlation is exactly the reason why discriminating between those who choose and those who do not choose $p_i = 1$ is beneficial for OP in the first place. Hence, OP's benefits from statistical discrimination are reduced by the chilling effect. This means that an evaluation of whether privacy should be given up will be biased against privacy if it does not take the behavior change of individuals, which also causes a reduction in the information gain for the OP, into account. The following proposition makes this statement more formally. To do so, we have to add the technical condition that the distribution Γ_0

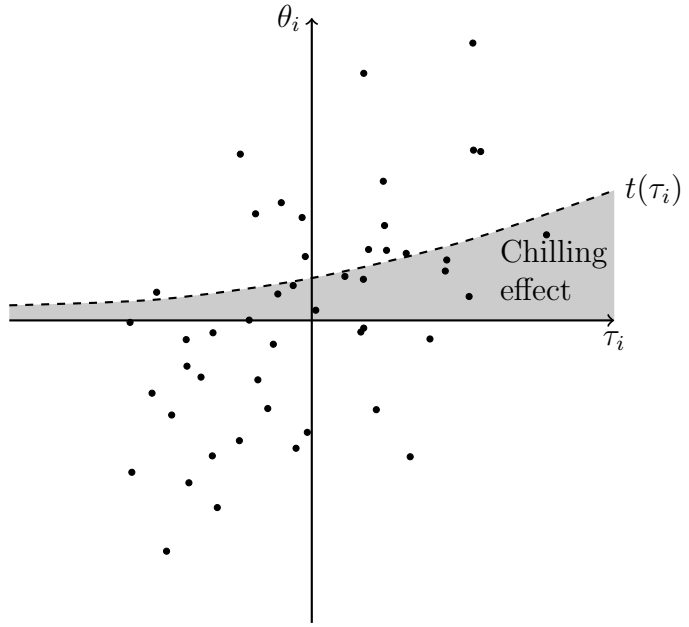

Figure 2: An illustration of proposition 1. If decisions are private, all individuals with positive θ_i will support $p = 1$ and all others support $p = 0$. If decisions are public, the OP can use the individuals' decisions to predict their type τ_i. Therefore, some people with relatively low θ_i will misrepresent their preferences to avoid the statistical discrimination. Since the disutility from being treated aggressively rises in τ_i, we get the curve above. Individuals in the gray area are subject to the chilling effect and support $p = 0$ without privacy.

is symmetric around 0. This ensures that OP does not gain from the fact that all cutoffs increase (while the argument above shows that it is detrimental to OP that cutoffs of higher τ increase by a larger amount). Note that the following proposition does not compare OP's payoffs under privacy and no privacy. In line with the argument above, it compares OP's payoffs without privacy with his payoffs in a hypothetical situation where there is no privacy but individuals use their equilibrium strategies of the privacy case.

Proposition 2 (Reduced information gain)**.** *Assume that the distribution $\Gamma_0(\tau)$ is symmetric around $\tau = 0$.*[11] *OP's payoff without privacy is lower if individuals use the cutoffs $t^{np}(\tau)$ than if they used the cutoffs $t^p(\tau) = 0$.*

As a side remark, note that the technical condition in proposition 2 is also sufficient to rule out somewhat uninteresting equilibria without privacy in which the OP plays M against everyone (i.e. equilibria in which the OP does not use the information he has): Given that Γ_0 is symmetric around zero, the OP would be indifferent between A and M if he knew that $\theta_i = 0$. By first order stochastic dominance, he will then prefer A to M when he knows that $\theta_i \geq 0$. But this is exactly the information $p_i = 1$ would give him because the cutoff in such a hypothetical equilibrium would clearly be the same as in the privacy case, namely zero. This contradicts that OP plays M against those choosing $p_i = 1$.

[11] An alternative technical condition that is also sufficient for the result to hold is $\mathbb{E}[\tau_i|\theta_i = 0] \geq 0$.

3. Welfare Analysis

What are the welfare consequences of the chilling effect? It is not hard to see that the chilling effect causes a welfare loss in the information aggregation stage. The bias against $p = 1$ means that information is no longer efficiently aggregated and decision 0 is more likely to be taken than optimal. The following lemma states formally that the privacy equilibrium yields a higher expected consumer surplus in the information aggregation stage than the equilibrium without privacy. (We define consumer surplus in the information aggregation stage as $p \sum_{i=1}^{n} \theta_i$.)

Lemma 3. *The cutoff strategy $t^p(\tau) = 0$, i.e. the equilibrium strategy in the privacy case, gives a higher expected consumer surplus in the information aggregation stage than any $t^{np}(\tau) > 0$.*

While this shows that individuals are always better off under privacy, this does not allow us to say anything about overall welfare. Without privacy, the OP can adjust his behavior according to people's choices p_i and thereby make use of the correlation between θ_i and τ_i to identify individuals with a relatively high τ_i.

Concerning overall welfare, we will derive three results that establish sufficient conditions for when privacy is welfare-optimal both for the individuals and the OP. Firstly, if the OP plays a mixed strategy in equilibrium (i.e. he mixes between treating people who choose $p_i = 1$ mildly or aggressively), privacy always provides higher welfare than no privacy. This simply follows from the fact that while individuals always lose from lack of privacy, the OP is indifferent between privacy and no privacy if he plays a mixed strategy in the no privacy case.

Secondly, we show that for large n, i.e. if there are many individuals, there exist no equilibria in which the OP plays a pure strategy, and privacy is therefore optimal.

Thirdly, we show the same for large $\delta(\tau)$ – in other words, if the cost of being treated aggressively is very high, then privacy is also welfare optimal. Figure 3 illustrates the proposition.

The second and the third part of the following proposition require the additional assumption that δ is strictly (and not just weakly) increasing in τ. This guarantees that we can make statements about how the correlation between p_i and τ_i develops in the limit.

Proposition 3 (Welfare comparison)**.** *1.) If OP uses a mixed strategy in the equilibrium without privacy, then privacy maximizes welfare.*
Assume $\delta' > 0$:
2.) For n sufficiently large, privacy welfare dominates no privacy in the following sense: Compared to the no privacy case, privacy leads to a higher expected consumer surplus and the same expected payoff for OP.
3.) Let the disutility of an individual facing action A by OP be $r\delta(\tau)$ (instead of $\delta(\tau)$). For r sufficiently large, privacy welfare dominates no privacy.

The intuition behind the second result is that the chilling effect is getting very large if the number of individuals grows. If n is large, each individual only has a small influence on the outcome of the information aggregation stage, while his "punishment" $\delta(\tau_i)$ from choosing $p_i = 1$ is independent of n. This implies that pure strategy separating equilibria no longer exist. Put differently, if OP played A against individuals who chose $p_i = 1$ and played M against individuals who played $p_i = 0$, then all individuals have an incentive to choose $p_i = 0$ in order to avoid the aggressive reaction by OP. With a low number of individuals, this incentive is countervailed by the downside of playing $p_i = 0$ (when θ_i is positive), since this makes the result $p = 0$ more likely. As n grows large, every individual's impact on this decision is, however, negligible and this negative effect cannot deter them from biasing their stated opinion. OP would then get no information at all from observing p_i, which contradicts his equilibrium strategy of playing A against $p_i = 1$. OP will, therefore, use a mixed strategy in equilibrium, which makes playing $p_i = 1$ less painful and therefore preserves some informativeness in the individuals' decisions. Hence, OP will be indifferent between his two actions, i.e. he would be equally well off by choosing M against everyone which would give him a payoff equal to his equilibrium payoff with privacy. Consequently, OP is indifferent between privacy and no privacy. Since individuals are clearly worse off without privacy because of the biased information aggregation and the possibly increased probability of A in the interaction stage, the privacy case welfare dominates.

The intuition for the third result is similar: If δ is high, then the benefit from the information aggregation stage is relatively small compared to the potential losses in the interaction stage. Individuals will therefore be chilled a lot if OP plays A against individuals who chose $p_i = 1$. Playing A for sure against those who chose $p_i = 1$ is then no longer a best response. Consequently, OP uses a mixed strategy for r sufficiently high and privacy welfare dominates.

Note that all the welfare results in proposition 3 are Pareto results *from an ex ante point of view*. That is, privacy makes individuals strictly better off in expectation (i.e. before knowing their type) while the OP is indifferent.

4. Alternative Utility Specifications

In this section, we consider two alternatives to the information aggregation in the first stage modeled so far. First, we consider a setup where individual i's utility does not depend on choices of other individuals. That is, the first stage decision p_i has nothing to do with information aggregation but is simply a private decision without externalities. Second, we consider a setting in which there is again information aggregation but individual i's payoff from $p = 1$ is given by a common state θ (instead of a personal payoff parameter θ_i). This state, however, is unknown, and individuals obtain only noisy private

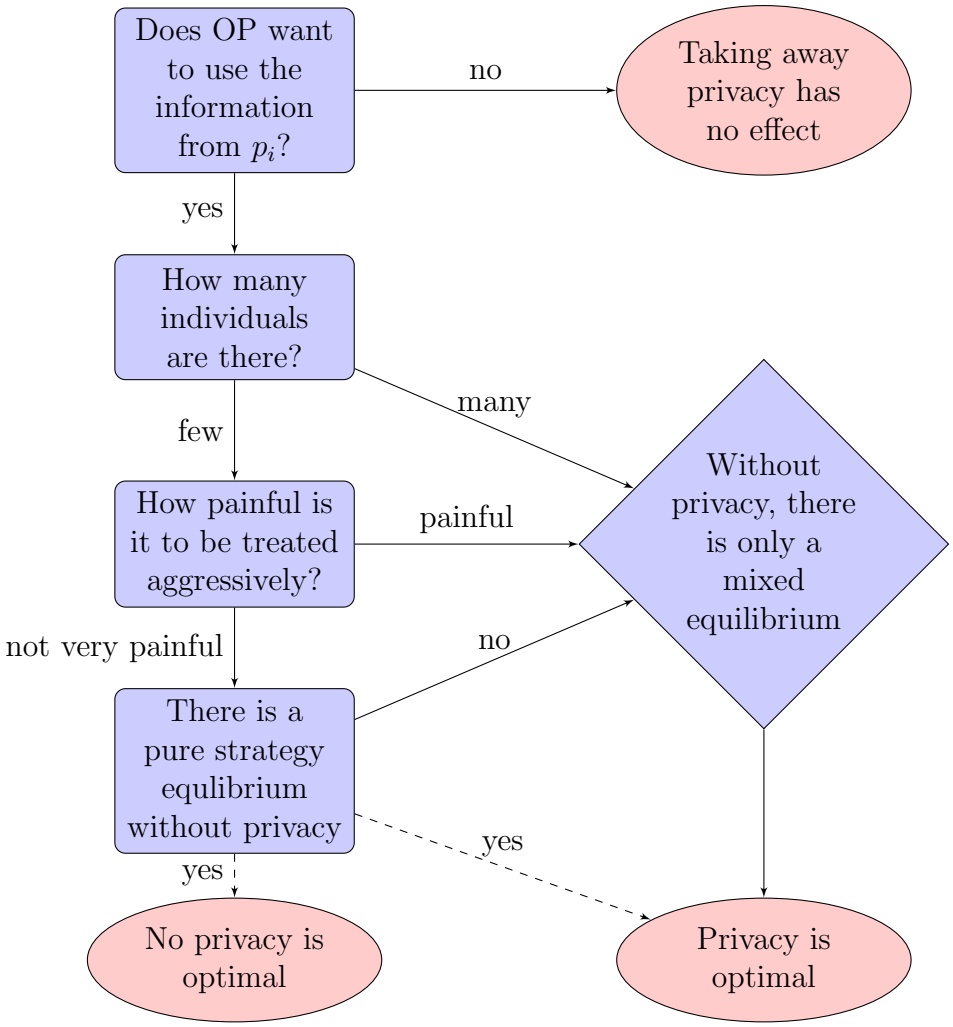

Figure 3: Sufficient conditions for when privacy is welfare-optimal, which follow from proposition 3. Note that the existence of a pure-strategy equilibrium without privacy is only a necessary, but not sufficient condition for privacy to be suboptimal. In section 4.1, we further investigate when privacy is optimal even in situations with pure equilibria in a special case of our model.

signals of the true state θ. As we will see, similar results to the ones above hold in these setups and some additional insights can be obtained.

4.1. First Stage With Private Decisions Instead of Information Aggregation

We want to consider a setup where individual i's choice (p_i) directly influence his welfare. Our model covers this latter case if we set $n = 1$. This could be the case for people listening to music, attending certain events or meeting certain people, which also is informative about some hidden type. Then p_i has a private consequence, and all of our results (about the existence of the chilling effect) continue to hold. But we can no longer argue that as each individual becomes less and less pivotal with larger n, the chilling effect increases and ultimately makes the statistical discrimination ineffective (proposition 3). In our example, this welfare question could be: If a preference for Reggae music is correlated with drug

use, should the employer be able to observe, and base his decision on, the music that Alice listens to?

Instead of concentrating on $n = 1$, we will simply assume that every individual's choice directly influences her payoff, so that her payoff from choosing p_i is simply $p_i\theta_i$. That is, we model n individuals but their welfare is independent from one another.[12] In the privacy case, preferences are the same as before. Without privacy, individuals experience a chilling effect that now only depends on the behavior of the OP and the function $\delta(\tau)$, and no longer on their beliefs about the behavior of others. As in proposition 2, the chilling effect can reduce the correlation and make the option of no privacy less attractive for the OP.

Example 2. *To give an example, consider the case of data-based police work. The purchase of precision scales through the online retailer Amazon suggests that the buyer might be a drug dealer: the predictive algorithm that suggests other items based on what people usually buy together with the scale are almost all drug-related.*[13] *Should the police (the OP in our model) be allowed to access Amazon's purchase data? From the outset, it might seem that this could help them track down drug dealers. If, however, purchase data was used in this way, it is clear that most drug dealers would be the first to procure their high precision scales in another way, and the police would be left with visiting a few enthusiastic coin collectors. The chilling effect would render the infringement of privacy useless.*

If the OP plays a mixed strategy in the equilibrium without privacy, we can show just as in proposition 3 above that privacy is Pareto optimal, since people are better off with privacy and the OP is indifferent. The interesting case is the one in which the OP uses a pure strategy, i.e. plays A against everybody who chooses $p_i = 1$ and M otherwise.

In this case, the OP is better off without privacy (otherwise he would always play M), while individuals are on average worse off. To be able to say anything about welfare, we need to aggregate their respective payoffs.

We can do so on an individual-by-individual basis, that is, we ask: For a given individual, what does this individual lose by losing privacy, and what does the OP gain? This is in line with viewing the OP as a representative of society (for example, the police trying to catch criminals or terrorists), or thinking of the OP as being a group of other players, or even of every individual acting as OP to another individual.

In the case where the OP uses a pure strategy, we can therefore write welfare as[14]

$$\sum_i p_i(\theta_i + \tau_i - \delta(\tau_i)).$$

[12] As this is equivalent to having n of our original models with one individual each, it is clear that our earlier intermediate results, i.e. lemmas 1–3, still hold.

[13] See https://www.cnet.com/uk/news/buy-a-scale-on-amazon-and-it-thinks-youre-a-drug-dealer/, retrieved September 29, 2016.

[14] We could use weights to sum up payoffs, but since we have made no assumptions about the magnitude of δ, a scaled δ function would just be another δ function and nothing qualitative would change.

When is welfare higher without privacy than with privacy? Intuitively, if the correlation between θ_i and τ_i is very high: then the OP's gain from being able to distinguish individuals according to type is also large, while the individual's loss from not being able to choose their preferred p_i (or being treated aggressively if they do) only depends on $\delta(\tau_i)$. For a given δ, the correlation between θ_i and τ_i would therefore have to be sufficiently high to make no privacy welfare-optimal. Figure 4 illustrates this intuition.

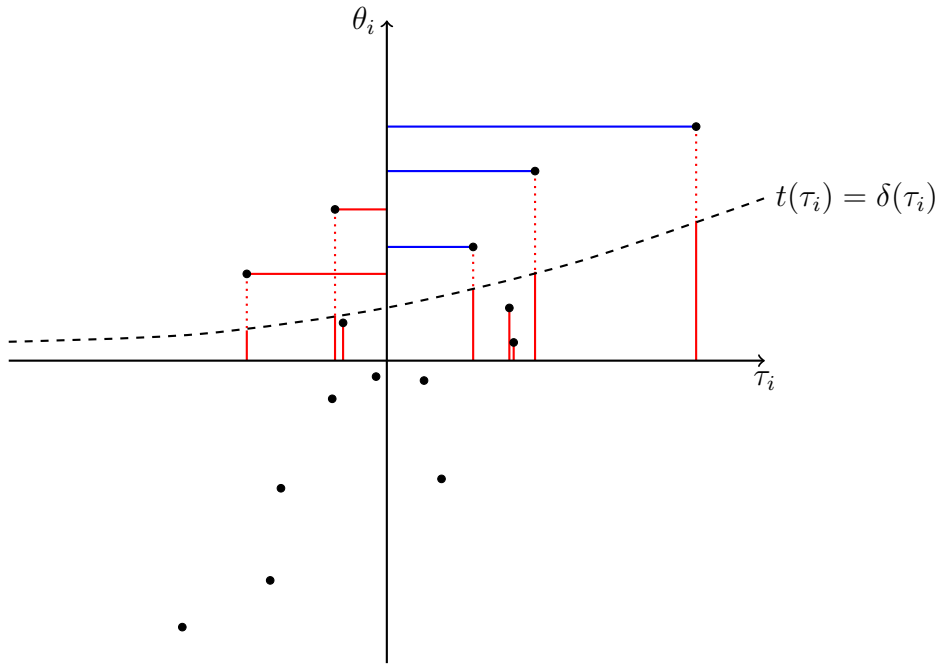

Figure 4: Gains (blue) and losses (red) from lack of privacy (compared to privacy). Losses of the individuals are vertical, losses and gains of the OP are horizontal. The sum of the lengths of all blue lines is the overall gain, the sum of all (solid) red lines is the overall loss. Every individual with $\theta_i > 0$ loses either θ_i (if she chooses $p_i = 0$) or $\delta(\tau_i)$ (if she chooses $p_i = 1$ and therefore gets treated aggressively). The OP gets τ_i for every individual who still chooses $p_i = 1$. Intuitively, if we increase correlation between θ_i and τ_i, individuals with $\theta_i > 0$ move to the right (as their expected τ_i increases), which increases the gain of the OP.

If we want to analyze the connection between correlation and δ, we need to restrict the problem by imposing partial orderings of joint distributions, since the set of possible joint distributions is otherwise intractable. We will therefore make our argument in two different ways. Firstly, we will consider the special case of $\delta(\tau) = \delta$ being a constant.[15] In this case, we can order distribution functions according to correlation based on first order stochastic dominance of Γ_θ. We then show that welfare is decreasing in δ and increasing in the correlation between θ_i and τ_i, and that for higher δ, no privacy is only optimal if the correlation is very high.[16] Secondly, we will consider the more general case

[15]Note that under this assumption, our limit results in proposition 3 are no longer valid. Since there is no information aggregation here, there can be no result for large n anyway.

[16]Of course, our result from above still applies: For very large δ, privacy is always welfare optimal.

of $\delta(\tau)$ being an increasing function while restricting the joint distribution of θ_i and τ_i to a family of distributions which are convex combinations of a correlated and an uncorrelated distribution. In both of these cases, we will show that for a given cost function for the individuals (i.e. a given δ), privacy is optimal both for the individuals and the OP unless the correlation between θ_i and τ_i is sufficiently high.

First, consider the case of constant δ. Recall that Γ_{θ_i} is the distribution of τ_i given θ_i; and that we have already assumed that $\Gamma_{\theta_i'}$ first-order stochastically dominates $\Gamma_{\theta_i''}$ if and only if $\theta_i' \geq \theta_i''$. Furthermore, we now assume that $\mathbb{E}\left[\tau_i|\theta_i = 0\right] \geq 0$ so that expected τ_i is positive for $\theta_i > 0$ – this guarantees that the OP wants to treat individuals aggressively if their θ_i is positive. We will now say that the correlation is higher in distribution Γ' than in distribution Γ'' if for every $\theta_i > 0$, Γ'_{θ_i} first-order stochastically dominates Γ''_{θ_i}. The following proposition shows that welfare is decreasing in δ and increasing in the correlation between θ_i and τ_i. This establishes that for a higher δ, the correlation between θ_i and τ_i needs to be higher to make no privacy welfare-optimal.

Proposition 4 (Monotone welfare difference)**.** *The welfare difference between no privacy and privacy is decreasing in δ and increasing in Γ.*

We now move to the second case of expressing "weak" correlation (and we allow again for increasing, i.e. non-constant, $\delta(\tau)$). We continue to focus on the interesting case in which – given distributions $\Gamma_{\theta_i}(\tau_i)$ – there is an equilibrium in which OP plays A (M) against those who choose $p_i = 1$ ($p_i = 0$). Now consider the marginal distribution of τ_i which we denote by $\bar{\Gamma}$, that is

$$\bar{\Gamma}(\tau_i) = \int_{\mathbb{R}} \Gamma_{\theta_i}(\tau_i)\, d\Phi(\theta_i).$$

If for every given θ_i the distribution of τ_i was $\bar{\Gamma}$, then there would be no correlation between θ_i and τ_i and even knowing θ_i directly (instead of p_i) would not yield any benefit for the OP. We will now consider convex combinations of the original distributions Γ_{θ_i} and the distribution $\bar{\Gamma}$. Denote these convex combinations by

$$\Gamma^{\lambda}_{\theta_i}(\tau_i) = \lambda\Gamma_{\theta_i}(\tau_i) + (1-\lambda)\bar{\Gamma}(\tau_i) \qquad \lambda \in [0,1].$$

For $\lambda = 1$ we are in the original problem. Decreasing λ, however, continuously decreases the correlation between θ_i and τ_i. For $\lambda = 0$, there is no correlation between these two variables left. If there is no correlation, then the equilibrium is the same as in the privacy case because OP does not get any information about τ_i from the choice of the individuals. Hence, the equilibrium is that OP plays M against everyone and individuals use the cutoff 0 if $\lambda = 0$. This is true regardless of whether there is privacy or not. By continuity, the same is true for low but positive λ. More interestingly, we establish in the following proposition that there is an intermediate range of λ where privacy is strictly

welfare superior to no privacy. That is, if correlation is weak privacy welfare dominates (and if the correlation is very weak privacy and no privacy are welfare equivalent).

Proposition 5 (Welfare optimality depending on type correlation)**.** *There exist* $0 < \underline{\lambda} < \bar{\lambda} < 1$ *such that*

1. *for* $\lambda \leq \underline{\lambda}$ *privacy and no privacy are welfare equivalent and*
2. *for* $\lambda \in (\underline{\lambda}, \bar{\lambda}]$ *privacy leads to strictly higher welfare than no privacy. The equilibrium for* $\lambda = \bar{\lambda}$ *is in pure strategies.*

4.2. Individuals with Identical Preferences Under Uncertainty

This subsection considers an alternative model where the private information of individuals in the information aggregation stage is not directly their personal payoff of $p = 1$. Instead, individuals all have the same payoff of $p = 1$ but each one of them only receives a noisy signal of this payoff. That is, there is an unknown state of the world θ; each individual has a noisy signal about the state of the world and they try to "match the state", i.e. they prefer $p = 1$ if the state is positive and $p = 0$ if the state is negative.

This has a striking implication: Lack of privacy makes *every* individual worse off, since the chilling effect inhibits information aggregation. In our main model, individuals have private preferences over outcomes and therefore some of them (those with negative θ_i) gain from the chilling effect. Since all individuals now have the same interest – implementing the policy that matches the state – everyone loses from the chilling effect's impact on information aggregation. Hence, our welfare result for large n in proposition 3 is somewhat stronger in this setting as privacy is now a Pareto improvement not only at the ex ante but even at the interim stage, i.e. after each individual has observed his signal.

The details of the setting are as follows: The state of the world θ is distributed standard normally and this θ is the payoff consequence of $p = 1$ for each individual. However, the realization of θ is unknown. Each individual obtains a private signal θ_i which is normally distributed around the true state θ, i.e. $\theta_i \sim N(\theta, \sigma^2)$ where we denote the cdf by $\tilde{\Phi}_\theta$ and the pdf by $\tilde{\phi}_\theta$. All θ_i are assumed to be independent draws from this distribution. The interaction type τ_i of individual i is drawn from Γ_{θ_i} where again $\Gamma_{\theta_i'}$ is assumed to first order stochastically dominate $\Gamma_{\theta_i''}$ if and only if $\theta_i' > \theta_i''$. This creates a positive correlation between θ_i and τ_i. The interaction stage is exactly the same as in our main model. That is, without privacy a strategy for OP states which of the two actions (A and M) OP plays against an individual who chose $p_i = 0$ or $p_i = 1$. With privacy, OP only decides which of the two actions he chooses against all individuals. This means that – to keep the setting comparable to the main model – we do not consider strategies (or beliefs) that are contingent upon the number of individuals choosing $p_i = 1$. This is a simplification. However, one can easily imagine settings where OP has to commit to a strategy before he gets to know the individuals' p_is. This is the case, for example, if the interaction is

between i and an agent representing OP and p_i is only learned in the interaction. OP then has to instruct the agent in advance how to act.

In the supplementary material to this paper, we provide proofs that are mostly analogous to those of our main model. In particular, the absence of privacy causes a chilling effect and this chilling effect inhibits efficient information aggregation. For large n, there are still only equilibria where the OP mixes, and in any equilibrium where the OP mixes, the OP is indifferent between privacy and no privacy. Now, however, we have Pareto dominance in the sense that every individual of every type is better off under privacy.

5. Extensions

This section contains three extensions to the main model. First, we relax the assumption that the information aggregation rule q is the same in the privacy and the no privacy case by allowing some hypothetical planner, who can be interpreted as a social norm or convention, to ex ante choose a q for each case in order to maximize the expected benefit from information aggregation. Second, we show that privacy might have to be mandated, i.e. privacy as an opt in possibility will lead to the no privacy outcome. Third, we consider the possibility of a defensive action against the OP and use this setup to show that in some scenarios privacy can even make the OP strictly better off.

5.1. An Endogenous Information Aggregation Process q

In this section we will endogenize the function q that assigns to each m/n a probability of implementing $p = 1$. In particular, we will assume that this function q is chosen by a planner in order to maximize the surplus in the information aggregation stage. The planner takes into account that individuals are chilled in the no privacy case and therefore the optimal q will differ in the privacy and no privacy case. The goal of this section is to show that our results from sections 2 and 3 remain valid in this setting.

We will assume that the planner has to choose an increasing function q and this function depends on the case – privacy and no privacy. For simplicity of exposition, we will assume that n is odd which ensures that a majority rule is clearly defined. Since we dropped the assumption that q is *strictly* increasing, we will require individuals to choose a cutoff strategy (after observing the case and q) $t(\tau)$ and we will concentrate on equilibria where each individual chooses the same cutoff strategy. If several such equilibria exist, we allow the planner to select the one that maximizes payoffs in the information aggregation stage. Consider the privacy case first. For any increasing q, it is a best response by the individuals to choose cutoff $t^p(\tau) = 0$. Given the independence of the θ_i, the planner's optimization problem is then

$$\max_q \sum_{m=0}^{n} q(m/n) \left(m\mathbb{E}[\theta_i|\theta_i \geq 0] + (n-m)\mathbb{E}[\theta_i|\theta_i < 0]\right).$$

As θ_i is normally distributed, $\mathbb{E}[\theta_i|\theta_i \geq 0] = -\mathbb{E}[\theta_i|\theta_i < 0]$ and it is easy to see that the optimal q is a majority rule, i.e. $p = 1$ if more than $n/2$ individuals choose $p_i = 1$ and $p = 0$ otherwise.

Now suppose for a moment that the planner could choose both t and q with the goal of maximizing expected surplus in the information aggregation stage. Given the symmetry of our setup, one can show that the planner would then choose $t = 0$ and majority rule; see the supplementary material for more details. That is, the privacy case delivers the maximal possible payoff in the information aggregation stage.

Next we consider the no privacy case. Obviously, a constant q is not optimal and therefore each individual's p_i will – with some probability – influence the decision on p. As in the main model, individuals will still be chilled to some extent if the OP's behavior depends on p_i, i.e. $t_q^{np}(\tau) \geq 0$ with strict inequality if the OP's behavior depends on p_i. Note that the threshold t_q^{np} depends on q. The planner's problem is now

$$\max_q \sum_{m=0}^{n} q(m/n) \left(m\mathbb{E}[\theta_i|\theta_i \geq t_q^{np}(\tau_i)] + (n-m)\mathbb{E}[\theta_i|\theta_i < t_q^{np}(\tau_i)] \right).$$

Given that $t_q^{np} \geq 0$ and that all θ_i are normally distributed, $q(1) = 1$ and $q(0) = 0$ are clearly optimal. All other values cannot be determined in general, that is, without specifying Γ_θ. Fortunately, our welfare result in proposition 3 can be derived without precise knowledge of q. The main argument is that the probability with which an individual expects to influence the decision on p is bounded from above by $1/n$. This follows directly from the assumption that all n individuals use the same strategy. Consequently, the same forces as in the original model are at work (regardless of the specific q): As n increases each individual is less likely to be pivotal and therefore the main motivation in the p_i choice is to avoid aggressive treatment by OP. The threshold t_q^{np} becomes arbitrarily high and steep and – due to the same reasoning as in the proof of proposition 3 – only mixed strategy equilibria exist. This implies that the OP is indifferent between privacy and no privacy. As mentioned above, the privacy case maximizes the expected payoff from information aggregation and therefore privacy welfare dominates no privacy.

5.2. Privacy as Opt In

Suppose that each individual has an additional decision to make in the information aggregation stage: They do not only have to choose p_i but also have to decide whether their choice should be private or public. OP can observe all public choices but not the private ones – in this case he can only observe that the individual chose privacy. To isolate the effect of the privacy choice, we will also assume that OP cannot make his behavior contingent on the outcome p (which might be realized only at a later point of time).

The possibility of hiding one's choice gives rise to multiple equilibria. To see this, consider first an equilibrium in which every individual always chooses "public" (no matter

what θ_i, τ_i or p_i is). Then the equilibrium of the case without privacy results.[17] Second, consider an equilibrium in which every individual always chooses "private". This means that we are effectively in the case with privacy. OP's best response is to play M and consequently no individual has an incentive to deviate.

Naturally, the question arises which of the two equilibria is more robust. We will argue in two different ways that the "always private" equilibrium is not very robust. The reason is an unraveling logic. Individuals who choose $p_i = 0$ are not afraid of making this public as it suggests that their θ_i is low, which means that their expected τ_i is also relatively low because of the positive correlation between the two. Given that the expected τ_i is low, OP would therefore still play M against those who make a choice $p_i = 0$ public. If, however, everyone who chooses $p_i = 0$ makes this public, then making one's choice private is not different from publicly choosing $p_i = 1$.

The simplest way to formalize this intuition is to assume that making one's choice p_i private comes at a small cost $\varepsilon > 0$. In this case, the "all private" equilibrium would only be supported by off equilibrium beliefs such that both $\mathbb{E}[\tau|\text{"public"}, p_i = 0] \geq 0$ and $\mathbb{E}[\tau|\text{"public"}, p_i = 1] \geq 0$ as OP could then threaten to play A against anybody making his decision public (thereby saving the $\varepsilon > 0$ costs). Given that $\mathbb{E}[\tau] < 0$, these are straightforwardly unreasonable beliefs. In terms of equilibrium refinements, the equilibrium does not satisfy the well known D1 criterion of Banks and Sobel (1987). Roughly speaking, this refinement states the following for our game: Denote by $D(\theta_i, \tau_i)$ the set of OP mixed strategies that are (i) best responses for some OP belief and (ii) would make a deviation by an individual of type (θ_i, τ_i) profitable. D1 requires that OP's off path beliefs must be zero for type (θ_i', τ_i') if there is a type (θ_i'', τ_i'') such that $D(\theta_i', \tau_i')$ is a strict subset of $D(\theta_i'', \tau_i'')$. Put differently, when facing an off-path deviation, OP should believe that it is more likely to be committed by a type whose deviation could be justified by a bigger set of OP beliefs. It is straightforward to show that the "all private" equilibrium does not satisfy D1. The reason is that the off path beliefs supporting the "all private" equilibrium require that deviations to public stem from individuals with relatively high τ_i no matter whether p_i is zero or one. As δ is increasing in τ_i, there are mixed strategies by OP which would make the deviation profitable for individuals with low τ_i (who are less afraid of action A) but not for individuals with high τ_i. The "all public" equilibrium, on the other hand, satisfies D1.

The second way in which the "all private" equilibrium is not robust is the following. Assume that with probability $\varepsilon > 0$ OP has alternative preferences $\tau_i + \varepsilon'$ for playing A. Assume that ε' is such that $\mathbb{E}[\tau] + \varepsilon' > 0$. That is, under the alternative preferences OP plays A given his prior beliefs. Suppose further that these alternative preferences are such that $\mathbb{E}[\tau|\theta_i \leq 0] + \varepsilon' < 0$, i.e. knowing that θ_i is negative OP still best responds

[17] This equilibrium is supported by the following off equilibrium path belief: if a player chooses "private", OP believes that τ_i is sufficiently high so that A is a best response.

by playing M. Again the "always private" equilibrium could then only be sustained by off path beliefs leading to $\mathbb{E}[\tau|\text{"public"}, p_i = 0] + \varepsilon' \geq 0$ and $\mathbb{E}[\tau|\text{"public"}, p_i = 1] + \varepsilon' \geq 0$. As pointed out above, such beliefs are unreasonable and violate the D1 refinement.

5.3. Defensive Actions

Suppose that individuals have the opportunity to take a defending action against being treated aggressively. More precisely, an individual can take an action D which increases his payoff if OP plays A but decreases his payoff if OP plays M. The defensive action reduces OP's payoff. In our example, Alice could hire a lawyer. Hiring the lawyer is costly but the lawyer will make it harder for the employer to discriminate against Alice. For the employer, dealing with a lawyer is a hassle (whether he discriminates or not) and reduces his payoffs.

What we want to show in this section is that the model can easily be extended in this way and that privacy could lead to (i) OP being *strictly* better off with privacy while (ii) individuals being in expectation strictly better off with privacy. Hence, privacy can be strictly Pareto superior from an ex ante point of view. To this end, it is sufficient to present an example and this is what we are going to do. Suppose $\tau_i \in \{\underline{\tau}, \bar{\tau}\}$, that is, τ_i can have only one of two values. Furthermore, assume that the probability that $\tau_i = \bar{\tau}$ equals

$$\gamma_{\theta_i} = \begin{cases} 0.7 - \frac{0.3}{\theta_i + 1} & \text{if } \theta_i \geq 0 \\ 0.1 - \frac{0.3}{\theta_i - 1} & \text{if } \theta_i < 0. \end{cases}$$

That is, the probability of a high $\bar{\tau}$ is increasing in θ_i and is point symmetric around $(0, 0.4)$. We take $\underline{\tau} = -2$, $\bar{\tau} = 3$ and $\delta(\tau_i)$ as given as in table 2.

action/type	$\underline{\tau}$	$\bar{\tau}$
not D	-0.1	-0.125
D	0.0	-0.025

Table 2: $-\delta(\tau_i)$ depending on whether the defensive action is taken.

If an individual takes action D and OP plays M, his payoff is -0.1, that is, the costs of the action D are 0.1. Note that the individual wants to play D if the chance of playing A is higher than 1/2. OP's payoffs are reduced by 1 if an individual plays D (for simplicity the payoff reduction is assumed to be independent of OP's action).

Under privacy, it is an equilibrium that every individual chooses $p_i = 1$ if and only if $\theta_i \geq 0$ while in the second stage OP plays M and no individual takes the action D. Without privacy, this is no longer an equilibrium as OP prefers to deviate by playing A against all individuals choosing $p_i = 1$: The probability that an individual is of type $\tau_i = \bar{\tau}$ given $\theta_i \geq 0$ (and therefore $p_i = 1$) is

$$\frac{\int_0^\infty \gamma_{\theta_i} \, d\Phi(\theta_i)}{2} \approx 0.51$$

which implies that OP's best response is A.

Let the probability that alternative 1 is chosen in stage 1 given m individuals supporting it be $q(m/n) = m/n$. Then we get the following equilibrium in the case without privacy: Individuals use cutoff strategies characterized by cutoffs $t(\underline{\tau}) = 0$ and $t(\bar{\tau}) = N * 0.025$. In the second stage, those individuals that chose $p_i = 1$ will play D. OP plays A against all individuals that chose $p_i = 1$ and M otherwise. To see that this is an equilibrium, note that an individual of type $(\theta_i, \tau_i) = (0, \underline{\tau})$ is indeed indifferent between choosing $p_i = 0$ and not playing D, which gives a payoff of 0 as OP will play M, and $p_i = 1$ and playing D which also gives a payoff of 0 as OP will then play A. Similarly, an individual of type $(\theta_i, \tau_i) = (0.025N, \bar{\tau})$ is indifferent between choosing $p_i = 0$ and not playing D and choosing $p_i = 1$ and playing D. The reason is that choosing $p_i = 1$ increases the probability of $p = 1$ being chosen by $1/N$ and therefore increases the expected payoff of an individual with $\theta_i = 0.025N$ by 0.025. However, the down side of choosing $p_i = 1$ is that the payoff in the interaction stage is 0.025 lower as $-\delta(\bar{\tau}) = -0.025$ (when playing D). For OP, the probabilities

$$
\begin{aligned}
prob(\tau_i = \bar{\tau}|p_i = 0) &= \frac{\int_{-\infty}^{0.025N} \gamma_{\theta_i}\, d\Phi(\theta_i)}{\int_{-\infty}^{0.025N} \gamma_{\theta_i}\, d\Phi(\theta_i) + \int_{-\infty}^{0} 1 - \gamma_{\theta_i}\, d\Phi(\theta_i)} \\
prob(\tau_i = \bar{\tau}|p_i = 1) &= \frac{\int_{0.025N}^{\infty} \gamma_{\theta_i}\, d\Phi(\theta_i)}{\int_{0.025N}^{\infty} \gamma_{\theta_i}\, d\Phi(\theta_i) + \int_{0}^{\infty} 1 - \gamma_{\theta_i}\, d\Phi(\theta_i)}
\end{aligned}
$$

are such that playing A (M) against those that chose $p_i = 1$ ($p_i = 0$) is optimal, i.e. $prob(\tau_i = \bar{\tau}|p_i = 0) \leq 0.4 \leq prob(\tau_i = \bar{\tau}|p_i = 1)$, if $N \leq 22$.[18]

OP's expected payoffs in the equilibrium without privacy are $-0.043 * N$ while OP's profits with privacy are zero. Individuals are strictly worse off if they have $\theta_i > 0$: The reasons are (i) that some are chilled and therefore expect a lower payoff from the information aggregation in stage 1, (ii) those that are not chilled have to endure action A by OP (and have to bear the costs of the defensive action). Individuals with $\theta_i < 0$ benefit from the chilling of others as this chilling implies that their personally preferred alternative is more likely to be implemented. Note, however, that – by the symmetry of the setup – this only offsets the first negative effect on those with $\theta_i > 0$ (in expectation, e.g. behind the veil of ignorance). The second negative effect on those with $\theta_i > 0$ lowers the expected payoff of any individual.

6. Discussion

This section explains first why privacy could be bad for welfare in certain situations and then discusses three applications of the ideas in this paper.

[18] For $N > 22$, no pure strategy equilibria exist without privacy and OP will therefore be indifferent between privacy and no privacy – cf. proposition 3.

6.1. When is Privacy Bad?

So far, we have mostly concentrated on situations and conditions in which privacy is beneficial for society, since this is the main novelty of our paper. But our model, and the assumptions under which we have derived our main results, also allow us to identify conditions under which privacy is not welfare-optimal – under which, in other words, intrusions into privacy can be justified. In addition to the conditions that follow from our results in sections 3 and 4, we will briefly comment on two additional restrictions here.

Biased information aggregation: One of the assumptions of our model with information aggregation has been that the information aggregation mechanism q is unbiased (in technical terms: that q was point-symmetric). If this assumption is not fulfilled, q systematically prefers either of the two options 0 and 1. This can be the case, for example, in problems of status-quo bias, or voting mechanisms that give a lot of power to single actors such as unanimity voting. In this case, the information aggregation mechanism in itself is clearly not welfare-optimal from a narrow utilitarian point of view (even though it could of course be justified by considerations such as protection of minorities). If we stick to the utilitarian welfare perspective, however, we can easily see that our main results do not apply, since the behavior of individuals under privacy is not welfare optimal and can potentially be improved by distorting it.

However, privacy can only improve welfare in this case if it distorts behavior in the "right" direction. Assume, for example, that q is biased such that it chooses option 1 with probability (almost) one if $m/n > 0.1$, and chooses option 0 with probability (almost) one otherwise. Lack of privacy, by lowering the propensity of individuals to support 1, can improve the probability that option 0 is chosen in cases where, for example, 60% of individuals prefer 0. This does not work the other way around: If q has a similar bias towards 0, lack of privacy will exacerbate the situation by guaranteeing that 1 gets chosen in even fewer situations where it would be the welfare-maximizing choice.

Negative externalities from choices: We could think of situations where one of the choices that individuals can take is inherently more desirable from a welfare perspective. For example, if the OP is trying to distinguish between criminals and non-criminals, and criminals are also more likely to enjoy engaging in small-scale vandalism, then introducing video surveillance (to detect vandalism) will have the benefit of identifying some potential criminals (those who are not subject to the chilling effect) *and* of deterring vandalism (through the chilling effect). While such externalities add an additional complication to our model, we can accommodate them by assuming that they subtract a certain length from all vertical loss lines in graph 4 that end below the curve of $t(\tau_i)$ (potentially making them negative). Our main welfare results change accordingly. In our model with information aggregation, no privacy can now be optimal even in cases of mixed equilibria, if

the gain that results from the chilling effect is large enough. And for any given set of parameters, no privacy is optimal for a larger interval of correlation parameters in our model without information aggregation.

6.2. Information Aggregation and Sorting Among Individuals: Opinion Polls and the Secret Ballot

The main informational tension that underlies questions of welfare and privacy is between inducing individuals to reveal valuable information by promising them influence on an outcome, and the countervailing threat of using such information to discriminate among individuals.

Perhaps the most prominent example, and one where almost everyone's intuition will come down on the side of privacy at least some of the time, is voting. Democratic societies use elections as a tool to collect information about their citizens' values, opinions, beliefs and preferences. It seems intuitively clear that the information aggregation in such elections can usually not be improved by making public how individuals have voted, as this would allow partisans of a candidate (the OP of our model) to intimidate or reward voters who would otherwise express themselves freely. In our main model, we derive sufficient conditions for when the information aggregation of a secret ballot cannot be improved by removing secrecy, and we further develop this result in section 5.1.

The effect of privacy can also be observed in the problem of predicting elections with opinion polls. Polls are usually conducted by interview and therefore offer less privacy than actual elections, and so respondents may adjust their answers to what they think the pollster wants to hear. This can be motivated by a fear of actual reprisals, or simply of being viewed unfavorably by the pollster conducting the interview – both are equivalent to the δ of our model.[19] At the same time, opinion polls offer only very limited influence to anyone who answers them, so that the δ can easily outweigh the benefit of answering honestly. The resulting bias in polls towards more socially acceptable options has become known under different names, such as the "Bradley effect" or the "Shy Tory factor".[20]

For a classic example of systematic error in opinion polls, consider the 1990 presidential election in Nicaragua, in which opinion polls varied widely and mispredicted the result substantially. Bischoping and Schuman (1992), in a well-known experiment, deployed researchers who conducted opinion polls while subtly looking as if they supported one of the candidates. This was achieved by having the pollsters using pens showing the symbol of either of the candidates to fill out the poll. Polls who were thus "associated" with different parties produced different results. In particular, polls who were either neutral or visually associated with the incumbent were quite different from the election result, while polls that seemed to be associated with the challenger were much closer to it. This suggests

[19]"Being viewed unfavorably" might seem like a reduced-form reputation effect, but if it causes direct disutility there is no difference in how our model would cover it.

[20]Newer terms like "Brexit effect" or "silent Trump vote" suggest that the phenomenon persists.

that, without the privacy of the voting booth, respondents weighed the potential costs of revealing their opinion to an "OP", and that this substantially chilled their expressed preferences.

Of course, a crucial point in this example is that the elections in Nicaragua were widely expected to be fair and secret, so that the chilling effect in the polls was not likely to be reproduced in the actual election. If a chilling effect will occur in the election itself, a poll is more likely to predict the result if it induces an effect of a similar magnitude in the same direction.

6.3. Which Discrimination Should be Permitted: Credit Scores

A bank has to decide to whom to lend. Ideally, it would like to base its decision on the probability that a debtor will repay the loan, but this variable is not directly observable. Instead, the bank can rely on measures that indirectly predict default probability. There are several socioeconomic variables that are easily observed and correlated with default risk, such as national origin, race, gender, age, or place of residence. Using such variables to make credit decisions, and hence treat native-borns, whites or women differently solely because of their identity, is illegal in many countries. In the United States, for example, such "redlining" practices are explicitly outlawed by the Equal Credit Opportunity Act (ECOA) of 1974.

Imagine, however, that the bank starts looking for other pieces of data that can inform its decision and allow it to statistically discriminate among loan applicants. Two such pieces of information are the education level (which can easily be documented by the applicant) and the taste in music (which many millions of people reveal on websites like Facebook, last.fm and similar services). While the former is common practice, the latter is (on purpose) more speculative but not implausible: Facebook owns a patent on aggregating credit scores from the data it collects about its users, and there are many firms that claim to make use of big data to develop more accurate credit scores.[21] We would expect that a preference for some genres of hip hop, since it is correlated with socioeconomic status, can be highly predictive of default risk. The expressed music preference would then be the variable p_i that the bank uses to discriminate between people who do and those who don't get loans, and our model would consequentially predict a chilling effect in which some hip hop fans are held back in their freedom of expression, since they want to improve their credit rating. The individual loss (by not being able to express your own personality) is probably more substantial than the loss in information aggregation here, but it is a welfare loss nonetheless (cf. the results in section 4.1). Note also that those who care too much to be stifled in their music appreciation will find it harder to get a loan, regardless of whether that is justified by their actual creditworthiness or not.

[21] One of them, Zest Finance, advertises with the slogan: "All Data is Credit Data." (https://www.zestfinance.com/how-we-do-it.html, retrieved May 2, 2016.)

But what is most important is that fans of gangsta rap tend to be similar to each other in many ways, so that the use of innocuous (and predictive) music preference data allows the bank to discriminate based on ethnicity, age and geography without explicitly saying so. This points to a larger question to which our research contributes, but to which we have no definitive answer: What should banks, employers, governments be allowed to discriminate upon? Most people would probably agree that to treat someone better or worse purely because of race or gender is not acceptable (and that contrary to the arguments made by Friedman (1962), such discrimination will not automatically disappear as it can be rational, as pointed out by Arrow (1973) and Phelps (1972)). But demanding that job applicants have a diploma, or giving loans based on past income, is also statistical discrimination: these factors are predictive of whether the employee will be up to the task or the loan will be repaid, but the correlation is less than 1.

Our first extension suggests that an equilibrium where everyone keeps their music preferences secret is not stable (especially if there is some payoff to sharing them). Regulation which prohibits the use of some data for credit decisions, beyond existing laws like the ECOA, could therefore be welfare-enhancing. In particular, recall that our model only requires that some variables are correlated without being causally related. Beside music taste, many other variables are probably correlated with both creditworthiness and race or gender without having any causal relationship with either of these. Clever bankers, or even mindless machine learning algorithms, could pick up on those relationships and use them to improve their credit algorithms, with all the consequences that we have described in our analysis. Our results would therefore strongly support the regulatory use of positive lists, which specify which data can be legally used in credit decisions (as opposed to negative lists, which only specify which data cannot be used).

6.4. "The Tape Has Had Some Chilling Effect": Decision-Making and Transparency

The last decades have seen a move towards transparency in many public bodies – governments, authorities, central banks. But to the extent that the quality of decisions in these institutions depends on aggregating the information of their employees and members, our results suggest that transparency does not necessarily improve welfare.

Consider, for example, the board of a central bank that has to decide on an interest rate change. If the deliberations are private and no minutes are made public, board members express their opinion quite freely.[22] If minutes are later published, however, members will worry about the effect of what they say on their reputation. Assume that board members have different degrees of competence, and that the probability of being wrong about something decreases in one's competence. Board members want to be thought of as competent by the public, their academic colleagues or future employers. Now, a board

[22]This is under our standard assumption that arguing one's viewpoint increases the probability that one's preferred policy will be implemented.

member considers whether to make an unconventional suggestion. This suggestion has some probability of being wrong, and in that case outside observers would adjust their belief of the board member's competence downwards. Publicity can therefore induce him to stay quiet.

This is in line with the results by Meade and Stasavage (2008) and others who examine the effects of a reform introduced in 1993, which mandated that minutes from meetings of the Federal Open Markets Committee (FOMC) of the U.S. Federal Reserve should be published. The reform has significantly increased conformity in the discussion and decreased the number of people who criticized the chairman's proposed interest rate adjustment. There was a strong shift away from free discussion and towards the reading of prepared statements. Thomas Hoenig, president of the Federal Reserve Bank of Kansas City, remarked in a meeting in 1995 that "the tape has had some chilling effect on our discussions. I see a lot more people reading their statements." (Meade and Stasavage, p. 13). Alan Greenspan also warned of this development before the reform was implemented: The FOMC "could not function effectively if participants had to be concerned that their half-thought-through, but nonetheless potentially valuable, notions would soon be made public."(Meade and Stasavage, 2008, p. 12)

Our model therefore suggests that if board members, government ministers or civil servants are worried about how they are being perceived by the outside world, secret meetings can substantially increase the quality of decision making.[23] But this is not universally true, of course. If the correlated types of our model reflect "private interests" (for example, stock holdings by family members) and "policy preferences", privacy would allow the board members to follow their private interests without having anything to fear – which would not improve decision making in the public interest. Privacy is not a panacea, but neither is transparency.

7. Conclusion

Why should an individual care about his or her privacy, why should a society care about the privacy of its members? We have argued that since asymmetric information is a fact of life, questions of privacy are never about whether there should be private information or not, but only how much there should be and how it should be structured. That allows us to answer: Individuals can worry that information about them could be used "against them", i.e. expose them to discriminative treatment. This result does not require ill will among the discriminator – the discrimination can be perfectly rational, as in the case of the employer trying to distinguish applicants. But it will make it harder for people to choose according to their preferences, and the rational reaction of individuals to having no privacy can impair the ability of a society to efficiently aggregate information. Privacy

[23] Consider also the literature on reputation concern and advice, such as Ottaviani and Sørensen (2006), which would also suggest that advisers are more helpful if they are unconcerned about their reputation.

is not only individually optimal, but also welfare-enhancing.

Our examples show, however, that privacy is not a silver bullet. The solution to problems of "redlining" and new forms of discrimination in lending is not to prohibit borrowers from revealing any information about themselves; and not all governments would be improved by being able to work in total secrecy. Our analysis allows us to say, however, when privacy is likely to improve welfare. When people's preferred actions under privacy guarantee an outcome that is optimal or close to optimal, the chilling effect decreases welfare. This is the case, for example, when actions have no significant externalities, or when the gains from correctly aggregating information are large.

Apart from the welfare effects, privacy often has a distributive effect: In our main model, there are always people whose preferred policy becomes less likely to be implemented under privacy. (In section 4.2, however, we argue that there can be situations where privacy improves everybody's outcome.) Others gain: Those who would be subject to the chilling effect without privacy are more likely to get their preferred option with privacy. Moreover, those with strong preferences gain twice from privacy: They are no longer statistically discriminated against, and their preferred option is more likely to be implemented. How should such distributive effects influence whether privacy is implemented? We have no definitive answer, but would like to point out that similar distributive effects arise with free speech: On any single issue, everybody would prefer if those with opposing viewpoints were prohibited from expressing it. Yet in the abstract, most of us would agree that freedom of expression should be universal.

We started this paper by criticizing the "Chicago view" of privacy as being inefficient and economically undesirable. But as we have argued that privacy can be fundamental to allowing individuals to freely express themselves, we are returning to an argument by perhaps the most well-known Chicago theorist. Friedman (1962, p. 52), in his discussion of "rules instead of authorities", discusses the question of whether free speech issues should be decided from case to case, or in the abstract. He concludes that:

> When a vote is taken on whether Mr. Jones can speak on the corner, it cannot allow [...] for the fact that a society in which people are not free to speak on the corner without special legislation will be a society in which the development of new ideas, experimentation, change, and the like will all be hampered in a great variety of ways that are obvious to all.

Our analysis suggests that a similar argument can be made about privacy.[24]

[24]It has been pointed out to us that the whistleblower Edward Snowden drew a similar comparison between privacy and free speech in an online debate: "Arguing that you don't care about the right to privacy because you have nothing to hide is no different than saying you don't care about free speech because you have nothing to say." (https://www.reddit.com/r/IAmA/comments/36ru89/just_days_left_to_kill_mass_surveillance_under/crglgh2, retrieved on July 1, 2016.)

Appendix

Technical Results

Lemma 4. *Let Φ be the standard normal distribution. Then $\int_{ka-b}^{ka} d\Phi / \int_{ka}^{\infty} d\Phi$ diverges to infinity as $k \to \infty$ for a, $b > 0$.*

Proof of lemma 4: We concentrate on the right tail of the standard normal distribution. If for all $x \in [ka-b, ka]$ and some constant c we have that $\frac{\phi(x)}{\phi(x+b)} \geq c$, then is is also true that

$$\frac{\int_{ka-b}^{ka} d\Phi}{\int_{ka}^{ka+b} d\Phi} \geq c.$$

(This can be seen by noting that the first inequality holds for the range of the integrals of the second inequality.) The pdf of the standard normal distribution is

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2},$$

and the quotient of $\phi(x)$ and $\phi(x+b)$ is therefore $e^{-\frac{1}{2}(x^2-(x+b)^2)} = e^{xb+\frac{1}{2}b^2}$. For $x \to \infty$, this quotient diverges, and hence $\frac{\int_{ka-b}^{ka} d\Phi}{\int_{ka}^{ka+b} d\Phi}$ diverges for $k \to \infty$. Now note that $\int_{ka}^{\infty} d\Phi = \int_{ka}^{ka+b} d\Phi + \int_{ka+b}^{ka+2b} d\Phi + \ldots$ and that for large k, the quotient between any summand on the RHS and the following summand diverges. This means that the overall sum is smaller than $2\int_{ka}^{ka+b} d\Phi$ as – for k sufficiently high – $\int_{ka}^{ka+b} d\Phi + \int_{ka+b}^{ka+2b} d\Phi + \ldots \leq \int_{ka}^{ka+b} d\Phi \sum_{i=0}^{\infty} (1/2)^i = 2\int_{ka}^{ka+b} d\Phi$. Since we have established above that $\frac{\int_{ka-b}^{ka} d\Phi}{\int_{ka}^{ka+b} d\Phi}$ diverges for large k, that means that $\frac{\int_{ka-b}^{ka} d\Phi}{\int_{ka}^{\infty} d\Phi}$ diverges as well. □

Proofs

Proof of lemma 1: For $\theta_i > (\max_{\tau_i} \delta(\tau_i)) / (min_k\{q(k/n) - q((k-1)/n) : k \in \{1, \ldots, n\}\})$, it is a dominant action to choose $p_i = 1$. Similarly, for $\theta_i < -(\max_{\tau_i} \delta(\tau_i)) / (min_k\{q(k/n) - q((k-1)/n)$ $k \in \{1, \ldots, n\}\})$, it is a dominant action to choose $p_i = 0$. Write the expected utility difference of playing $p_i = 1$ and playing $p_i = 0$ as[25]

$$-\delta(\tau_i)\Delta + \theta_i * \sum_{k=1}^{n} ((q(k/n) - q((k-1)/n)) * prob(k-1)) \tag{5}$$

[25] In principle Δ could depend on the number of individuals choosing $p_i = 1$ in the information aggregation stage. In this case, the expected utility difference is

$$\sum_{k=1}^{n} \{[-\delta(\tau_i)\Delta(k, k-1) + \theta_i] ((q(k/n) - q((k-1)/n)) * prob(k-1))\}$$

where $\Delta(k, k-1)$ is the difference between the believed probability that OP plays A when facing an individual who played $p_i = 1$ and k individuals chose 1 and the probability that OP plays A when facing an individual who played $p_i = 0$ and $k-1$ individuals chose 1. The same argument as below holds: this expression is strictly increasing in θ_i. As will become apparent from (1)– (4), OP's best response strategy will not depend on the number of individuals choosing 1; see the comment in footnote 10.

where $prob(k-1)$ is i's belief that exactly $k-1$ others will choose $p_j = 1$ and $\Delta \in [-1, 1]$ is the difference between the (believed) probability that OP plays A when facing an individual who has played $p_i = 1$ and an individual who has played $p_i = 0$. Clearly, (5) is strictly increasing and continuous in θ_i. As it is optimal to play $p_i = 1$ ($p_i = 0$) if (5) is positive (negative), the best response to any given belief is a cutoff strategy where the cutoff is given by the θ_i for which the utility difference above is 0. (Note that the dominance regions above establish that an interior cutoff exists.) Since all best responses are cutoff strategies, all rationalizable actions are cutoff strategies.

In the privacy case, $\Delta = 0$ by definition and therefore (5) is zero if and only if $\theta_i = 0$ as the sum is clearly positive (recall that the cumulative distribution function q was strictly increasing by assumption). Consequently, $t^p(\tau_i) = 0$. □

Proof of lemma 2: Suppose $v_1 < v_0$ in equilibrium. In this case, (5) is strictly increasing in τ_i as $\Delta < 0$ and therefore $t(\tau_i)$ is strictly decreasing in τ_i.

This implies that we can partition $\mathbb{R}$ in three intervals $(-\infty, t(\bar{\tau})]$, $(t(\bar{\tau}), t(\underline{\tau})]$, $(t(\underline{\tau}), \infty)$. Denoting the inverse of the equilibrium cutoff t by s, we get

$$\begin{aligned}
v_1 &= \frac{\int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{s(\theta_i)}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i) + \int_{t(\underline{\tau})}^{\infty} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{s(\theta_i)}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i) + \int_{t(\underline{\tau})}^{\infty} \int_{\underline{\tau}}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)} \\
&\geq \frac{\int_{t(\bar{\tau})}^{\infty} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{t(\bar{\tau})}^{\infty} \int_{\underline{\tau}}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)} \\
&> \frac{\int_{-\infty}^{t(\underline{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{-\infty}^{t(\underline{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)} \\
&\geq \frac{\int_{-\infty}^{t(\bar{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i) + \int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{\underline{\tau}}^{s(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{-\infty}^{t(\bar{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i) + \int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{\underline{\tau}}^{s(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)} \\
&= v_0
\end{aligned}$$

where the inequalities use the assumption that $\Gamma_{\theta_i'}$ first order stochastically dominates $\Gamma_{\theta_i''}$ if $\theta_i' > \theta_i''$ and therefore θ_i and τ_i are positively correlated.[26] The result that $v_0 < v_1$ contradicts our initial supposition and therefore $v_1 \geq v_0$ in all equilibria. □

[26] To be clear, take the first of the inequalities:

$$\begin{aligned}
\mathbb{E}[\tau|\theta_i > t(\bar{\tau})] = \frac{\int_{t(\bar{\tau})}^{\infty} \mathbb{E}[\tau|\theta_i] \, d\Phi(\theta_i)}{\int_{t(\bar{\tau})}^{\infty} \int_{\underline{\tau}}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)} &\leq \frac{\int_{t(\bar{\tau})}^{t(\underline{\tau})} \mathbb{E}[\tau|\theta_i] \int_{z(\theta_i)}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\theta_i + \int_{t(\underline{\tau})}^{\infty} \mathbb{E}[\tau|\theta_i] \, d\Phi(\theta_i)}{\int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{z(\theta_i)}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i) + \int_{t(\underline{\tau})}^{\infty} d\Phi(\theta_i)} \\
&\leq \frac{\int_{t(\bar{\tau})}^{t(\underline{\tau})} \mathbb{E}[\tau|\theta_i, \tau \geq z(\theta_i)] \int_{z(\theta_i)}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\theta_i + \int_{t(\underline{\tau})}^{\infty} \mathbb{E}[\tau|\theta_i] \, d\Phi(\theta_i)}{\int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{z(\theta_i)}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i) + \int_{t(\underline{\tau})}^{\infty} d\Phi(\theta_i)} = v_1
\end{aligned}$$

where the first inequality holds as $\mathbb{E}[\tau|\theta_i]$ is strictly increasing in θ_i (by the first order stochastic dominance assumption on Γ_{θ_i}) and therefore putting less weight on lower θ_i increases the expectation. The third inequality follows a similar logic and the second one uses that $\mathbb{E}[\tau|\theta_i]$ is strictly increasing in θ_i directly.

Proof of proposition 1: Consider (5) which has to be zero if θ_i equals the equilibrium cutoff level. By lemma 2, $\Delta \geq 0$. In an equilibrium of the privacy case $\Delta = 0$ by assumption and $t^p(\tau_i) = 0$, see lemma 1. In the case without privacy, $t^{np}(\tau_i) < 0$ is impossible as then both terms in (5) are negative (recall $\Delta \geq 0$) at $\theta_i = t^{np}(\tau_i)$ with the second term being strictly negative. Consequently, the two terms could not sum to zero. We can therefore conclude that $t^{np}(\tau_i) \geq 0$ which establishes $t^{np}(\tau_i) \geq t^p(\tau_i)$. Note that this inequality is strict if $\Delta^{np} > 0$ as (5) would not equal zero for $\theta_i = 0$ and $\Delta > 0$.

Next we have to show that $\Delta > 0$ whenever the equilibrium strategy of OP is influenced by the presence of privacy. By lemma 2, $\Delta \geq 0$. If OP behavior is influenced by the presence of privacy and $\Delta = 0$ then the probability of A has to change in both groups (individuals choosing $p_i = 0$ and individuals choosing $p_i = 1$) by the same amount compared to the privacy case. For concreteness, suppose the probability of A is increased. This implies that in the privacy case the probability of A is less than 1. Consider first the case that OP has strict preferences in the privacy equilibrium which then implies that OP played A with probability 0 in the privacy equilibrium. Consequently, $v_p = \int_{\mathbb{R}} \int_{\underline{\tau}}^{\bar{\tau}} \tau \; d\Gamma_{\theta_i} \; d\Phi(\theta_i) < 0$. As $\beta_0(\tau)$ and $\beta_1(\tau)$ are obtained by means of Bayesian updating, it is impossible that both $v_0 \geq 0$ and $v_1 \geq 0$. But then it is impossible that playing A against both groups with (the same) positive probability is optimal in the equilibrium without privacy. Second, consider the case where OP is indifferent in the privacy equilibrium and plays A with some probability $\alpha < 1$. Indifference means that $v_p = 0$. If $\Delta = 0$ in the case without privacy, then it is easy to see that $v_1 > v_0$ because of the positive correlation of τ_i and θ_i. But this would imply $v_1 > 0$ and $v_0 < 0$ and therefore $\Delta = 0$ would not be a best response which contradicts that $\Delta = 0$ in equilibrium in the second case. If $\Delta > 0$, however, we already established above that $t^{np}(\tau_i) > t^p(\tau_i)$. The proof for a decrease of the probability of playing A is analogous.

Last we show that the difference $t^{np}(\tau_i) - t^p(\tau_i)$ is increasing in τ_i. As mentioned above $t^p(\tau_i) = 0$. Using the fact that (5) has to be zero at the cutoff, we obtain

$$t^{np}(\tau_i) = \frac{\Delta}{\sum_{k=1}^{n}(q(k/n) - q((k-1)/n)) * prob(k-1)} \delta(\tau_i). \tag{6}$$

Note that $prob(k-1)$ is independent of individual i's type τ_i as these types are drawn independent from one another. Therefore, the only term in $t^{np}(\tau_i) - t^p(\tau_i)$ which depends on τ_i is $\delta(\tau_i)$ which is increasing by assumption. As the fraction on the right-hand side of (6) is positive, it follows that $t^{np}(\tau_i) - t^p(\tau_i)$ is increasing in τ_i. □

Proof of proposition 2: We start with the case where OP finds it optimal to play A against all individuals choosing $p_i = 1$ and M against all individuals choosing $p_i = 0$ under both strategies t^{np} and t^p. Recall that OP's payoff is the expected value of τ of all those individuals against which OP plays A. Hence, the payoff difference of OP's payoff between the two scenarios is the expected value of τ in the area between the horizontal

axis and t^{np} in figure 5 below.

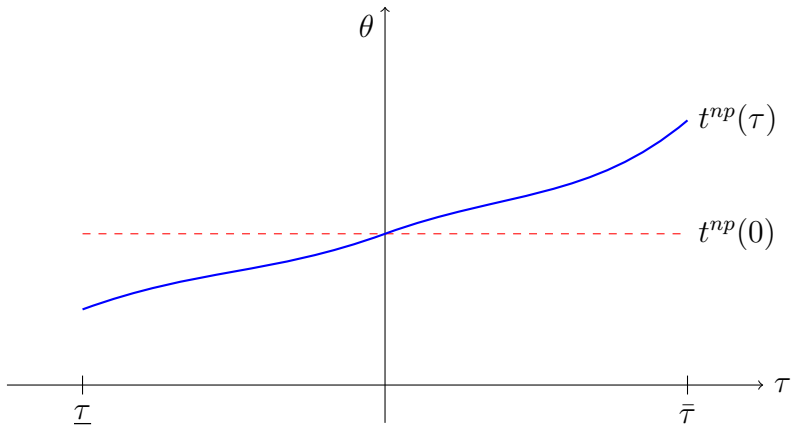

Figure 5: Integration range for difference in OP payoff

Denote the inverse function of $t^{np}(\tau)$ as $z(\theta)$. The difference of OP's payoffs between individuals using t^{np} and t^{p} is

$$\int_0^{t^{np}(\underline{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_\theta(\tau) d\Phi(\theta) + \int_{t^{np}(\underline{\tau})}^{t^{np}(\bar{\tau})} \int_{z(\theta)}^{\bar{\tau}} \tau \, d\Gamma_\theta(\tau) d\Phi(\theta)$$
$$= \int_0^{t^{np}(0)} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_\theta(\tau) d\Phi(\theta) - \int_{t^{np}(\underline{\tau})}^{t^{np}(0)} \int_{\underline{\tau}}^{z(\theta)} \tau \, d\Gamma_\theta(\tau) d\Phi(\theta) + \int_{t^{np}(0)}^{t^{np}(\bar{\tau})} \int_{z(\theta)}^{\bar{\tau}} \tau \, d\Gamma_\theta(\tau) d\Phi(\theta)$$

where the equality simply splits up the integration range which can be easily visualized in figure 5. The first of the three double integrals is positive by the following argument: As – by assumption – Γ_0 is symmetric around 0, $\int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_0(\tau) = 0$. It follows that $\int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_\theta(\tau) > 0$ for all $\theta > 0$ because Γ_θ first order stochastically dominates Γ_0 for all $\theta > 0$. This implies that the first double integral is positive as $t^{np}(0) \geq 0$ by proposition 1. The second double integral is negative as it integrates only over $\tau \leq 0$ and with the minus sign this second term becomes positive as well. The third double integral is positive as it integrates only over positive τ. Consequently, OP would like to play A against individuals with (τ_i, θ_i) in the area between the horizontal axis and t^{np} which means that OP is better off (given the strategy of playing A if and only if $p_i = 1$) under $t^p(\tau) = 0$ than under t^{np}.

We established that playing A against individuals who play $p_i = 1$ is relatively more attractive if individuals use strategy $t^p(\tau) = 0$ than if they use strategy t^{np}. This implies that whenever OP prefers to play A against individuals who play $p_i = 1$ under t^{np}, the same is true under t^p. Hence, we do not have to consider a case where OP plays M against individuals choosing $p_i = 1$ if they use t^p but A if they use t^{np}. In all other cases, OP uses the same action against individuals choosing $p_i = 0$ and against individuals choosing $p_i = 1$. Hence, $t^p = t^{np}$ and OP's payoffs are the same under both strategies (t^p and t^{np}). □

Proof of lemma 3: As the type draws are independent across individuals and as τ

is not payoff relevant in the information aggregation stage, it is clear that the consumer surplus optimal cutoff will be independent of τ. Suppose cutoff $t^* \geq 0$ is consumer surplus optimal. A necessary condition for optimality is the following: Say an individual has type $\theta_i = t^*$, then his choice must be consumer surplus neutral. That is, whether he chooses $p_i = 0$ or $p_i = 1$ must lead to the same expected consumer surplus (conditional on his own type being $\theta_i = t^*$). If this condition was not satisfied, either in- or decreasing t^* will increase expected consumer surplus thereby contradicting the optimality of t^*. We will show that the only t^* satisfying this necessary condition is $t^* = 0$.

The expected difference of consumer surplus when choosing $p_i = 1$ and $p_i = 0$ is (where we write t instead of t^* to shorten notation)

$$t+\sum_{l=0}^{n-1}\binom{n-1}{l}\Phi(t)^l(1-\Phi(t))^{n-1-l}(q((l+1)/n)-q(l/n))\left(l\mathbb{E}[\theta|\theta<t]+(n-1-l)\mathbb{E}[\theta|\theta>t]\right) \tag{7}$$

where l is the number of others choosing $p_i = 0$ (according to the cutoff strategy t). First, consider the term $l = (n-1)/2$ (in case n is odd). For this term $l = (n-1-l)$ and as $\mathbb{E}[\theta|\theta < t]+\mathbb{E}[\theta|\theta < t] \geq 0$ by $t \geq 0$, $p_i = 1$ will lead to a higher expected consumer surplus in this case. For $l < (n-1)/2$, we clearly have $l\mathbb{E}[\theta|\theta < t] + (n-1-l)\mathbb{E}[\theta|\theta < t] > 0$ by $t > 0$ and again $p_i = 1$ increases expected consumer surplus. However, for $l > (n-1)/2$ the opposite might be the case. Hence, we have to weigh terms with different l against each other. In particular, we will consider the terms $l > (n-1)/2$ and $n-1-l < (n-1)/2$ jointly. By the assumption that q is point symmetric around $1/2$, $q((l+1)/n) - q(l/n) = q((n-1-l+1)/n) - q((n-1-l)/n)$. Furthermore, the binomial coefficient is symmetric around the mean which means that also $\binom{n-1}{l} = \binom{n-1}{n-1-l}$. Consequently, we can write the sum of the two terms corresponding to l and $n-1-l$ as follows (using $z = 2l-n+1$ and dropping the argument of $\Phi(t)$ to save space)

$$\begin{aligned}&\binom{n-1}{l}\Phi^{n-1-l}(1-\Phi)^{n-1-l}(q((l+1)/n)-q(l/n))\\&\quad\left\{\mathbb{E}[\theta|\theta<t]\left(l\Phi^z+(n-1-l)(1-\Phi)^z\right)+\mathbb{E}[\theta|\theta>t]\left((n-1-l)\Phi^z+l(1-\Phi)^z\right)\right\}.\end{aligned}$$

We will now argue that the expression in curly brackets (and therefore the whole expression) is positive (for any $l > (n-1)/2$). Note that the only negative term in the curly brackets is $\mathbb{E}[\theta|\theta < t]$. Also recall that $t \geq 0$ and therefore $\Phi \geq 1-\Phi$. This implies that the term in curly brackets is (weakly) greater than

$$\mathbb{E}[\theta|\theta<t]\left((n-1)\Phi^z\right)+\mathbb{E}[\theta|\theta>t]\left((n-1)(1-\Phi)^z\right)$$

where we increased the weight on the negative term as much as possible (recall that $n-1 \geq l > (n-1)/2$). This means that the term in curly brackets is definitely positive

if $\mathbb{E}[\theta|\theta < t]\Phi^z + \mathbb{E}[\theta|\theta > t](1-\Phi)^z \geq 0$. Given that z is an integer between 1 and $n-1$, this inequality is hardest to satisfy for $z = 1$ but there it holds as $\mathbb{E}[\theta|\theta < t]\Phi(t) + \mathbb{E}[\theta|\theta > t](1-\Phi(t)) = 0$ by the definition of conditional expectations.

Now that we know that the expression in curly brackets is positive for all $l > (n-1)/2$ we can conclude that $p_i = 1$ leads to a higher expected consumer surplus than choosing $p_i = 0$ and this is true in a strict sense if $t > 0$: The analysis of the case $l = (n-1)/2$ and the consideration of twin pairs $l > (n-1)/2$ and $n-1-l$ have shown that the sum in (7) is positive and adding $t \geq 0$ keeps (7) positive (strictly if $t > 0$). It is easy to verify that the sum in (7) equals zero if $t = 0$ (in this case $\Phi = 1 - \Phi = 1/2$). This implies that $t = 0$ satisfies the necessary condition for optimality and all other $t > 0$ do not.[27]

Hence, $t = 0$ is optimal if we can verify that an optimal t exists. Note that expected consumer surplus is continuous in t. As it is straightforward that $t \to \infty$ is not optimal, the problem of finding the optimal t can be reduced to maximizing a continuous function over a compact set and a solution exists by the Weierstrass theorem. □

Proof of proposition 3: Let M be optimal for OP in the privacy equilibrium.

1.) Suppose there is a mixed strategy equilibrium in the case without privacy. Then, OP has to play M against both groups with positive probability. If he played A against those who chose $p_i = 1$ for sure and mixed for those who chose $p_i = 0$, then M could not be optimal in the privacy case. Hence, OP can in the case without privacy achieve a payoff equal to his equilibrium payoff by playing M against both groups. Consequently, OP's payoff with and without privacy is the same. Individuals are strictly better off with privacy as (a) there is no chilling effect which means by lemma 3 that expected welfare in the information aggregation stage is maximized and (b) M will be played with probability 1 against them in the interaction stage.

2.) Now assume that $\delta'(\tau) > 0$. We will show that for n sufficiently high the privacy equilibrium welfare dominates the equilibrium in the case without privacy (or the two are identical).

We are going to make use of the fact that for any $\mu > 0$, we can find an $\hat{n}$ so that for all $n > \hat{n}$, $q((m+1)/n) - q(m/n) < \mu$. This follows from the assumptions that q is strictly increasing and continuously differentiable. Intuitively, q would have to have an infinite slope somewhere for this not to be true.

Now consider (5) and suppose $\Delta = 1$. Note that $\sum_{j=1}^{n}((q(j/n) - q((j-1)/n)) * prob(j-1)) < \sum_{j=1}^{n} \mu * prob(j-1)) \leq \mu$, which gets arbitrarily small as n gets large. Consequently, the threshold values become arbitrarily large as n gets large. Note also that using (5) we can then write $t(\tau_i) = \delta(\tau_i)/(\sum_{j=1}^{n}((q(j/n) - q((j-1)/n)) * prob(j-1))) \geq \delta(\tau_i)/\mu$. Similarly, $t'(\tau_i) = \delta'(\tau_i)/(\sum_{j=1}^{n}((q(j/n) - q((j-1)/n)) * prob(j-1))) \geq \delta'(\tau_i)/\mu$. Hence, t is increasing in τ and the slope also becomes arbitrarily large as n increases (as μ can be chosen arbitrarily small for n sufficiently large).

[27] As the setup is symmetric, a similar argument could be made to rule out the optimality of any $t < 0$.

Denoting the inverse of the threshold t by z, we can write

$$v_1 = \frac{\int_{t(\underline{\tau})}^{t(\bar{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{1 - \Phi(t(\bar{\tau})) + \int_{t(\underline{\tau})}^{t(\bar{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)} + \frac{\mathbb{E}[\tau | \theta_i > t(\bar{\tau})]}{1 + \frac{\int_{t(\underline{\tau})}^{t(\bar{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{1 - \Phi(t(\bar{\tau}))}}.$$

As z becomes arbitrarily flat for n sufficiently high, we can choose – for n high enough – an $\varepsilon > 0$ such that $\int_{t(\bar{\tau})-\varepsilon}^{t(\bar{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)/(1-\Phi(t(\bar{\tau}))) > 0.5 \int_{t(\bar{\tau})-\varepsilon}^{t(\bar{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)/(1-\Phi(t(\bar{\tau})))$. It follows that the second term in v_1 goes to zero as $n \to \infty$ because $\int_{t(\bar{\tau})-\varepsilon}^{t(\bar{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)/(1 - \Phi(t(\bar{\tau})))$ and therefore its denominator diverges to infinity by lemma 4.

The first term in v_1 converges to something below the unconditional mean of τ which we denote by $\tau^E = \mathbb{E}[\tau]$: For n large, the previous step implies that,

$$\begin{aligned} v_1 \quad &\approx \quad \frac{\frac{\int_{t(\underline{\tau})}^{t(\tau^E)} \int_{\underline{\tau}}^{z(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{t(\underline{\tau})}^{t(\tau^E)} \int_{\underline{\tau}}^{z(\theta_i)} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)} + \frac{\int_{t(\tau^E)}^{t(\bar{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{t(\underline{\tau})}^{t(\tau^E)} \int_{\underline{\tau}}^{z(\theta_i)} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}}{1 + \frac{\int_{t(\tau^E)}^{t(\bar{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i) + 1 - \Phi(t(\bar{\tau}))}{\int_{t(\underline{\tau})}^{t(\tau^E)} \int_{\underline{\tau}}^{z(\theta_i)} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}} \\ &\leq \quad \frac{\int_{t(\underline{\tau})}^{t(\tau^E)} \int_{\underline{\tau}}^{z(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{t(\underline{\tau})}^{t(\tau^E)} \int_{\underline{\tau}}^{z(\theta_i)} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)} + \frac{\int_{t(\tau^E)}^{t(\bar{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{t(\underline{\tau})}^{t(\tau^E)} \int_{\underline{\tau}}^{z(\theta_i)} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)} \end{aligned}$$

Note that the first term equals $\mathbb{E}[\tau_i | t(\underline{\tau}) \leq \theta_i \leq t(\tau^E) \ \wedge \ \tau_i \leq z(\theta_i)]$. Clearly, this is below the unconditional mean τ^E. It follows that for a sufficiently small $\varepsilon' > 0$ (and large n)

$$v_1 \leq \tau^E + \frac{\int_{t(\tau^E)}^{t(\bar{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{t(\tau^E)-\varepsilon'}^{t(\tau^E)} \int_{\underline{\tau}}^{\bar{\tau}} \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}.$$

Note that the same ε' appropriately chosen for some n will also work for higher n (as the density of ϕ thins out for higher θ_i and $t(\tau^E) - t(\underline{\tau})$ is increasing in n). This implies that we can conclude for the limit $n \to \infty$ that

$$v_1 \leq \tau^E + \frac{\int_{t(\tau^E)}^{t(\bar{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\Phi(\theta_i)}{\int_{t(\tau^E)-\varepsilon'}^{t(\tau^E)} \int_{\underline{\tau}}^{\bar{\tau}} \, d\Gamma_{\infty}(\tau) \, d\Phi(\theta_i)} \leq \tau^E + \frac{\bar{\tau} \int_{t(\tau^E)}^{t(\bar{\tau})} \, d\Phi(\theta_i)}{\int_{t(\tau^E)-\varepsilon'}^{t(\tau^E)} \, d\Phi(\theta_i)} \xrightarrow{n \to \infty} \tau^E$$

where the limit follows from lemma 4 and the above established fact that t goes to infinity as $n \to \infty$. By assumption, OP's best response when facing the unconditional mean τ^E (or a lower τ_i) is M which contradicts the supposition $\Delta = 1$. Hence, $\Delta < 1$ which implies that OP uses a mixed strategy. By the first part of the proposition, privacy then welfare dominates no privacy.

3.) We will show that OP either plays M (independent of p_i) or uses a mixed strategy in the no privacy equilibrium if r is sufficiently high. (1) will then imply (3).

Suppose OP plays a pure strategy in equilibrium. If OP plays M against $p_i = 1$, then – by the assumption that OP plays M in the privacy case – privacy and no privacy case lead to the same equilibrium and the result holds trivially. OP cannot play A against $p_i = 0$: By lemma 2, OP would then also play A against $p_i = 1$. But this is incompatible with Bayesian updating and the assumption that OP plays M in the privacy case. Hence, we only need to consider the case where OP plays M against $p_i = 0$ and A against $p_i = 1$. Consider (5) which can be rearranged to get (under the assumption that $\Delta = 1$, i.e. OP plays A against $p_i = 1$ and M against $p_i = 0$)

$$t^{np}(\tau) = \frac{r\delta(\tau)}{\sum_{k=1}^{n}(q(k/n) - q((k-1)/n))\, prob(k-1)} \geq r\delta(\tau).$$

Hence, t^{np} diverges to ∞ as $r \to \infty$. Furthermore, the slope of t^{np} is linearly growing in r. Hence, the derivative of $t^{np}(\tau)$ also diverges to ∞ as r grows. But then the same steps as in the proof of result (2) above imply that $v_1 \leq \tau^E$, i.e. playing A against $p_i = 1$ is not a best response which contradicts that OP uses the pure strategy corresponding to $\Delta = 1$ in the equilibrium without privacy for r sufficiently large. As – for r sufficiently large – OP uses either mixed strategy in the no privacy equilibrium or plays M regardless of p_i, (1) implies that privacy dominates no privacy. □

Proof of proposition 4: The welfare-difference between no privacy and privacy is given by

$$\int_{\delta}^{\infty} \int_{\underline{\tau}}^{\overline{\tau}} \tau_i d\Gamma_{\theta_i} d\Phi(\theta_i) - \int_{\delta}^{\infty} \delta d\Phi(\theta_i) - \int_{0}^{\delta} \theta_i \Phi(\theta_i). \tag{8}$$

If we increase δ, it is clear that the first term weakly decreases, as the area of the integral gets smaller. To see what happens to the second and third term, we can disaggregate them further, assuming that we increase δ by ϵ. Then we get a net effect of

$$-\int_{\delta+\epsilon}^{\infty} \delta d\Phi - \int_{\delta+\epsilon}^{\infty} \epsilon d\Phi - \int_{0}^{\delta+\epsilon} \theta_i \Phi + \int_{\delta}^{\infty} \delta d\Phi + \int_{0}^{\delta} \theta_i \Phi$$
$$= -\int_{\delta+\epsilon}^{\infty} \epsilon d\Phi - \int_{\delta}^{\delta+\epsilon} (\theta_i - \delta) d\Phi,$$

which is negative. Therefore, the overall welfare decreases in δ.

Now consider what happens if we increase Γ. This only has influence on the payoff of the OP; the second and third term in (8) above remain unchanged. It follows from our definition that if we replace Γ' with Γ'' and $\Gamma' < \Gamma''$, then $\int_{\underline{\tau}}^{\overline{\tau}} \tau_i d\Gamma''_{\theta_i} > \int_{\underline{\tau}}^{\overline{\tau}} \tau_i d\Gamma'_{\theta_i}$. This means that if we increase Γ, the inner integral in the first term in (8) increases, and hence the whole term increases. □

Proof of proposition 5: First consider $\lambda = 0$. Note that the distribution of τ_i under $\bar{\tau}$ is the same as the distribution of τ_i that the OP faces in the privacy case of the original model (with distribution Γ_{θ_i}). As we assumed that OP plays M in the privacy

equilibrium, it is clear that the privacy equilibrium is also an equilibrium for $\lambda = 0$. In fact, it is the unique equilibrium: Since M is the best response against the distribution $\bar{\Gamma}$ by assumption, OP has to play M for sure against at least one group of individuals (either those choosing $p_i = 0$ or those choosing $p_i = 1$) by Bayesian updating. Suppose OP played A with positive probability against those who chose $p_i = 1$. Then some individuals with low θ_i would be chilled and play $p_i = 0$. As δ is increasing in τ_i, the best response cutoff would be increasing in τ_i, see (6). But then the average τ_i among those choosing $p_i = 1$ is lower than the average τ_i under $\bar{\Gamma}$. Consequently, M is a strict best response by OP because M is a best response against $\bar{\Gamma}$. This contradicts that OP plays A with positive probability.

Note that $\mathbb{E}[\tau_i|\theta_i \geq 0]$ is continuous in λ. Since M is a best response against $\bar{\Gamma}$, that is $\mathbb{E}[\tau_i|\theta_i \geq 0] < 0$ for $\lambda = 0$, the same is true for sufficiently small $\lambda > 0$. Hence, a $\underline{\lambda} > 0$ exists such that for all $\lambda \leq \underline{\lambda}$ the unique equilibrium without privacy is that OP plays M and all individuals use a cutoff of zero. This is equivalent to the privacy equilibrium and therefore privacy and no privacy are welfare equivalent for all $\lambda \leq \underline{\lambda}$. For the result in the proposition, let $\underline{\lambda}$ be the highest λ such that the equilibrium in the no privacy is that OP plays M against individuals choosing $p_i = 1$. Note that $\underline{\lambda} < 1$ as by assumption OP plays A against individuals choosing $p_i = 1$ for $\lambda = 1$.

For $\lambda = 1$, the equilibrium of the no privacy case was assumed to be that OP plays A (M) against $p_i = 0$ ($p_i = 1$) in the no privacy case. Denote by λ^* the infimum of all λ for which such an equilibrium exists. Clearly, $\lambda^* \in (\underline{\lambda}, 1)$. Since such an equilibrium no longer exists for $\lambda < \lambda^*$, it has to hold true that at $\lambda = \lambda^*$ OP is indifferent between playing A and playing M against those playing $p_i = 1$ (for lower λ OP will then prefer to play M as the correlation is too weak and that is why the equilibrium breaks down). Note that the best response cutoffs of the individuals do not depend on λ but only on the OP's strategy. It follows that $\mathbb{E}[\tau_i|\theta_i \geq t^{np}(\tau_i)]$ is continuous in λ for $\lambda \geq \lambda^*$. As the OP is indifferent at λ^*, we have $\mathbb{E}[\tau_i|\theta_i \geq t^{np}(\tau_i)] = 0$ at λ^*. Continuity, implies that $\mathbb{E}[\tau_i|\theta_i \geq t^{np}(\tau_i)]$ is arbitrarily small for λ close but strictly above λ^*. That is, for any $\varepsilon > 0$ there is a $\varepsilon' > 0$ such that imposing privacy leads only to less than ε losses for the OP if $\lambda < \lambda^* + \varepsilon'$. Imposing privacy leads (for $\lambda \in [\lambda^*, \lambda^* + \varepsilon']$) to a discrete increase in citizen welfare for several reasons: First, those choosing $p_i = 1$ no longer face the aggressive response which increases their payoff by $\delta(\tau_i)$. Second, in the privacy case individuals use the cutoff zero instead of $t^{np} > 0$ which leads to a higher surplus in the information aggregation stage. This implies that for $\varepsilon' > 0$ small enough, privacy welfare dominates no privacy for $\lambda \in (\lambda^*, \lambda^* + \varepsilon']$. Let $\bar{\lambda} = \lambda^* + \varepsilon'$. Note that for $\lambda \in (\underline{\lambda}, \bar{\lambda})$ the equilibrium in the no privacy case is necessarily mixed which means implies that privacy is Pareto dominant for these λ, see proposition 3. This establishes the claim. □

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